CHAPTER

6 The Stability of Linear Feedback Systems

PREVIEW

Stability of closed-loop feedback systems is central to control system design. A stable system should exhibit a bounded output if the input is bounded. This is known as bounded-input, bounded-output stability. The stability of a feedback system is directly related to the location of the roots of the characteristic equation of the system transfer function and to the location of the eigenvalues of the system matrix for a system in state variable format. The Routh–Hurwitz method is introduced as a useful tool for assessing system stability. The technique allows us to compute the number of roots of the characteristic equation in the right half plane without actually computing the values of the roots. This gives us a design method for determining values of certain system parameters that will lead to closed-loop stability. For stable systems, we will introduce the notion of relative stability which allows us to characterize the degree of stability. The chapter concludes with a stabilizing controller design based on the Routh–Hurwitz method for the Sequential Design Example: Disk Drive Read System.

DESIRED OUTCOMES

Upon completion of Chapter 6, students should:

- Understand the concept of stability of dynamic systems.
- Be aware of the key concepts of absolute and relative stability.
- Be familiar with the notion of bounded-input, bounded-output stability.
- Understand the relationship of the s-plane pole locations (for transfer function models) and of the eigenvalue locations (for state variable models) to system stability.
- Know how to construct a Routh array and be able to employ the Routh–Hurwitz stability criterion to determine stability.

6.1 THE CONCEPT OF STABILITY

When considering the design and analysis of feedback control systems, **stability** is of the utmost importance. From a practical point of view, a closed-loop feedback system that is unstable is of minimal value. As with all such general statements, there are exceptions; but for our purposes, we will declare that all our control designs must result in a closed-loop stable system. Many physical systems are inherently open-loop unstable, and some systems are even designed to be open-loop unstable. Most modern fighter aircraft are open-loop *unstable by design*, and without active feedback control assisting the pilot, they cannot fly. Active control is introduced by engineers to stabilize the unstable system—that is, the aircraft—so that other considerations, such as transient performance, can be addressed. Using feedback, we can stabilize unstable systems and then with a judicious selection of controller parameters, we can adjust the transient performance. For open-loop stable systems, we still use feedback to adjust the closed-loop performance to meet the design specifications. These specifications take the form of steady-state tracking errors, percent overshoot, settling time, time to peak, and the other indices.

We can say that a closed-loop feedback system is either stable or it is not stable. This type of stable/not stable characterization is referred to as **absolute stability**. A system possessing absolute stability is called a stable system—the label of absolute is dropped. Given that a closed-loop system is stable, we can further characterize the degree of stability. This is referred to as **relative stability**. The pioneers of aircraft design were familiar with the notion of relative stability—the more stable an aircraft was, the more difficult it was to maneuver (that is, to turn). One outcome of the relative instability of modern acrobatic aircraft is high maneuverability. A acrobatic aircraft is less stable than a commercial transport; hence it can maneuver more quickly. As we will discuss later in this section, we can determine that a system is stable (in the absolute sense) by determining that all transfer function poles lie in the left-half s-plane, or equivalently, that all the eigenvalues of the system matrix $\mathbf{A}$ lie in the left-half s-plane. Given that all the poles (or eigenvalues) are in the left-half s-plane, we investigate relative-stability by examining the relative locations of the poles (or eigenvalues).

A **stable system** is defined as a system with a bounded (limited) system response. That is, if the system is subjected to a bounded input or disturbance and the response is bounded in magnitude, the system is said to be stable.

A stable system is a dynamic system with a bounded response to a bounded input.

The concept of stability can be illustrated by considering a right circular cone placed on a plane horizontal surface. If the cone is resting on its base and is tipped slightly, it returns to its original equilibrium position. This position and response are said to be stable. If the cone rests on its side and is displaced slightly, it rolls with no tendency to leave the position on its side. This position is designated as the neutral stability.

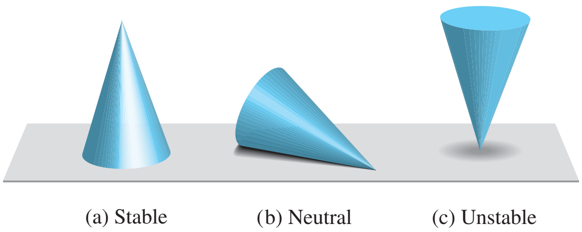


FIGURE 6.1 Illustration of stability.

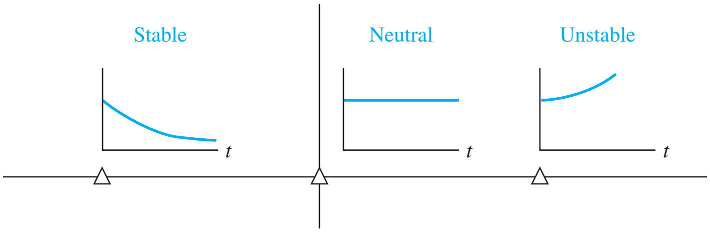


FIGURE 6.2 Stability in the s-plane.

On the other hand, if the cone is placed on its tip and released, it falls onto its side. This position is said to be unstable. These three positions are illustrated in Figure 6.1.

The stability of a dynamic system is defined in a similar manner. The response to a displacement, or initial condition, will result in either a decreasing, neutral, or increasing response. Specifically, it follows from the definition of stability that a linear system is stable if and only if the absolute value of its impulse response $g(t)$, integrated over an infinite range, is finite. That is, in terms of the convolution integral Equation (5.2) for a bounded input, $\int_0^\infty |g(t)|\, dt$ must be finite.

The location in the s-plane of the poles of a system indicates the resulting transient response. The poles in the left-hand portion of the s-plane result in a decreasing response for disturbance inputs. Similarly, poles on the $j\omega$-axis and in the right-hand plane result in a neutral and an increasing response, respectively, for a disturbance input. This division of the s-plane is shown in Figure 6.2. Clearly, the poles of desirable dynamic systems must lie in the left-hand portion of the s-plane [1–3].

A common example of the potential destabilizing effect of feedback is that of feedback in audio amplifier and speaker systems used for public address in auditoriums. In this case, a loudspeaker produces an audio signal that is an amplified version of the sounds picked up by a microphone. In addition to other audio inputs, the sound coming from the speaker itself may be sensed by the microphone. The strength of this particular signal depends upon the distance between the loudspeaker and the microphone. Because of the attenuating properties of air, a larger distance will cause a weaker signal to reach the microphone. Due to the finite propagation speed of sound waves, there will also be a time delay between the signal produced by the loudspeaker and the signal sensed by the microphone. In this case, the output from the feedback path is added to the external input. This is an example of positive feedback.

As the distance between the loudspeaker and the microphone decreases, we find that if the microphone is placed too close to the speaker, then the system will be unstable. The result of this instability is an excessive amplification and distortion of audio signals and an oscillatory squeal.

In terms of linear systems, we recognize that the stability requirement may be defined in terms of the location of the poles of the closed-loop transfer function. A closed-loop system transfer function can be written as

$$T(s) = \frac{p(s)}{q(s)} = \frac{K\prod_{i=1}^{M}(s + z_i)}{s^N \prod_{k=1}^{Q}(s + \sigma_k)\prod_{m=1}^{R}[s^2 + 2\alpha_m s + (\alpha_m^2 + \omega_m^2)]}, \tag{6.1}$$

where $q(s) = \Delta(s) = 0$ is the characteristic equation whose roots are the poles of the closed-loop system. The output response for an impulse function input (when $N = 0$) is then

$$y(t) = \sum_{k=1}^{Q} A_k e^{-\sigma_k t} + \sum_{m=1}^{R} B_m \left(\frac{1}{\omega_m}\right) e^{-\alpha_m t} \sin(\omega_m t + \theta_m), \tag{6.2}$$

where A_k and B_m are constants that depend on σ_k, z_i, α_m, K, and ω_m. To obtain a bounded response, the poles of the closed-loop system must be in the left-hand portion of the s-plane. Thus, **a necessary and sufficient condition for a feedback system to be stable is that all the poles of the system transfer function have negative real parts**. A system is stable if all the poles of the transfer function are in the left-hand s-plane. A system is not stable if not all the roots are in the left-hand plane. If the characteristic equation has simple roots on the imaginary axis ($j\omega$-axis) with all other roots in the left half-plane, the steady-state output will be sustained oscillations for a bounded input, unless the input is a sinusoid (which is bounded) whose frequency is equal to the magnitude of the $j\omega$-axis roots. For this case, the output becomes unbounded. Such a system is called **marginally stable**, since only certain bounded inputs (sinusoids of the frequency of the poles) will cause the output to become unbounded. For an unstable system, the characteristic equation has at least one root in the right half of the s-plane or repeated $j\omega$ roots; for this case, the output will become unbounded for any input.

For example, if the characteristic equation of a closed-loop system is

$$(s + 10)(s^2 + 16) = 0,$$

then the system is said to be marginally stable. If this system is excited by a sinusoid of frequency $\omega = 4$, the output becomes unbounded.

An example of how mechanical resonance can cause large displacements occurred in a 39-story shopping mall in Seoul, Korea. The *Techno-Mart* building, shown in Figure 6.3, hosts activities such as physical aerobics, in addition to shopping. After a Tae Bo workout session on the 12^{th} floor with about twenty participants, the building shook for 10 minutes triggering an evacuation for two days [5]. A team of experts concluded that the building was likely excited to mechanical resonance by the vigorous exercise.

To ascertain the stability of a feedback control system, we could determine the roots of the characteristic polynomial $q(s)$. However, we are first interested in determining the answer to the question, Is the system stable? If we calculate the roots of the characteristic equation in order to answer this question, we have determined much more information than is necessary. Therefore, several

FIGURE 6.3 Vigorous exercising on the 12th floor likely led to mechanical resonance of the building triggering a two-day evacuation. (Photo courtesy of Truth Leem/Reuters.)

methods have been developed that provide the required yes or no answer to the stability question. The three approaches to the question of stability are (1) the s-plane approach, (2) the frequency ($j\omega$) approach, and (3) the time-domain approach.

Industrial robot sales were the highest level ever recorded for a single year in 2013. In fact, since the introduction of industrial robots at the end of the 1960s until 2013, there have been over 2.5 million operational industrial robots sold. The worldwide stock of operational industrial robots at the end of 2013 was in the range of 1.3–1.6 million units. The projections are that from 2015–2017, industrial robot installations will increase by 12% on average per year [10]. Clearly, the market for industrial robots is dynamic. The worldwide market for service robots is similarly active. The projections for the period 2014–2017 are that approximately 31 million new service robots for personal use (such as vacuum cleaners and lawn mowers) and approximately 134,500 new service robots for professional use will be put into service [10]. As the capability of robots increases, it is reasonable to assume that the numbers in service will continue to rise. Especially interesting are robots with human characteristics, particularly those that can walk upright [21]. The IHMC robot depicted in Figure 6.4 competed in the recent DARPA Robotics Challenge [24]. Examining the IHMC robot in Figure 6.4, one can imagine that it is not inherently stable and that active control is required to keep it upright during the walking motion. In the next sections we present the Routh–Hurwitz stability criterion to investigate system stability by analyzing the characteristic equation without direct computation of the roots.

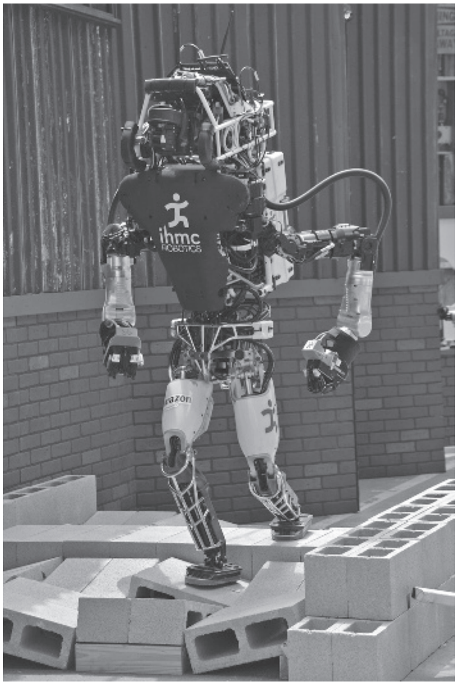


FIGURE 6.4
Team *IHMC* on the rubble on the first day of the DARPA Robotics Challenge 2015.
(Photo courtesy of DARPA.)

6.2 THE ROUTH–HURWITZ STABILITY CRITERION

The discussion and determination of stability has occupied the interest of many engineers. Maxwell and Vyshnegradskii first considered the question of stability of dynamic systems. In the late 1800s, A. Hurwitz and E. J. Routh independently published a method of investigating the stability of a linear system [6, 7]. The Routh–Hurwitz stability method provides an answer to the question of stability by considering the characteristic equation of the system. The characteristic equation is written as

$$\Delta(s) = q(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0. \tag{6.3}$$

To ascertain the stability of the system, it is necessary to determine whether any one of the roots of $q(s)$ lies in the right half of the s-plane. If Equation (6.3) is written in factored form, we have

$$a_n(s - r_1)(s - r_2)\cdots(s - r_n) = 0, \tag{6.4}$$

where $r_i = i$th root of the characteristic equation. Multiplying the factors together, we find that

$$\begin{aligned} q(s) = a_n s^n &- a_n(r_1 + r_2 + \cdots + r_n)s^{n-1} \\ &+ a_n(r_1 r_2 + r_2 r_3 + r_1 r_3 + \cdots)s^{n-2} \\ &- a_n(r_1 r_2 r_3 + r_1 r_2 r_4 \cdots)s^{n-3} + \cdots \\ &+ a_n(-1)^n r_1 r_2 r_3 \cdots r_n = 0. \end{aligned} \tag{6.5}$$

In other words, for an nth-degree equation, we obtain

$$\begin{aligned} q(s) = a_n s^n &- a_n \text{ (sum of all the roots) } s^{n-1} \\ &+ a_n \text{ (sum of the products of the roots taken 2 at a time) } s^{n-2} \\ &- a_n \text{ (sum of the products of the roots taken 3 at a time) } s^{n-3} \\ &+ \cdots + a_n (-1)^n \text{ (product of all } n \text{ roots)} = 0. \end{aligned} \tag{6.6}$$

Examining Equation (6.5), we note that all the coefficients of the polynomial will have the same sign if all the roots are in the left-hand plane. Also, it is necessary that all the coefficients for a stable system be nonzero. These requirements are necessary but not sufficient. That is, we immediately know the system is unstable if they are not satisfied; yet if they are satisfied, we must proceed further to ascertain the stability of the system. For example, when the characteristic equation is

$$q(s) = (s+2)(s^2 - s + 4) = (s^3 + s^2 + 2s + 8), \tag{6.7}$$

the system is unstable, and yet the polynomial possesses all positive coefficients.

The **Routh–Hurwitz criterion** is a necessary and sufficient criterion for the stability of linear systems. The method was originally developed in terms of determinants, but we shall use the more convenient array formulation. The Routh–Hurwitz criterion is based on ordering the coefficients of the characteristic equation

$$a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \cdots + a_1 s + a_0 = 0 \tag{6.8}$$

into an array as follows [4]:

$$\begin{array}{c|cccc} s^n & a_n & a_{n-2} & a_{n-4} & \cdots \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \end{array}$$

Further rows of the array, known as the Routh array, are then completed as

$$\begin{array}{c|cccc} s^n & a_n & a_{n-2} & a_{n-4} & \cdots \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\ s^{n-2} & b_{n-1} & b_{n-3} & b_{n-5} & \cdots \\ s^{n-3} & c_{n-1} & c_{n-3} & c_{n-5} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \\ s^0 & h_{n-1} & & & \end{array}$$

where

$$b_{n-1} = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} = \frac{-1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix},$$

$$b_{n-3} = -\frac{1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix}, \ldots$$

$$c_{n-1} = \frac{-1}{b_{n-1}} \begin{vmatrix} a_{n-1} & a_{n-3} \\ b_{n-1} & b_{n-3} \end{vmatrix}, \quad \ldots$$

and so on. The algorithm for calculating the entries in the array can be followed on a determinant basis or by using the form of the equation for b_{n-1}.

The Routh–Hurwitz criterion states that the number of roots of $q(s)$ with positive real parts is equal to the number of changes in sign of the first column of the Routh array. This criterion requires that there be no changes in sign in the first column for a stable system. This requirement is both necessary and sufficient.

Four distinct cases or configurations of the first column array must be considered, and each must be treated separately and requires suitable modifications of the array calculation procedure: (1) No element in the first column is zero; (2) there is a zero in the first column, but some other elements of the row containing the zero in the first column are nonzero; (3) there is a zero in the first column, and the other elements of the row containing the zero are also zero; and (4) as in the third case, but with repeated roots on the $j\omega$-axis.

To illustrate this method clearly, several examples will be presented for each case.

Case 1. No element in the first column is zero.

EXAMPLE 6.1 Second-order system

The characteristic polynomial of a second-order system is

$$q(s) = a_2 s^2 + a_1 s + a_0.$$

The Routh array is written as

$$\begin{array}{c|cc} s^2 & a_2 & a_0 \\ s^1 & a_1 & 0 \\ s^0 & b_1 & 0 \end{array},$$

where

$$b_1 = \frac{a_1 a_0 - (0)a_2}{a_1} = \frac{-1}{a_1}\begin{vmatrix} a_2 & a_0 \\ a_1 & 0 \end{vmatrix} = a_0.$$

Therefore, the requirement for a stable second-order system is that all the coefficients be positive or all the coefficients be negative. ■

EXAMPLE 6.2 Third-order system

The characteristic polynomial of a third-order system is

$$q(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & a_3 & a_1 \\ s^2 & a_2 & a_0 \\ s^1 & b_1 & 0 \\ s^0 & c_1 & 0 \end{array},$$

where

$$b_1 = \frac{a_2 a_1 - a_0 a_3}{a_2} \quad \text{and} \quad c_1 = \frac{b_1 a_0}{b_1} = a_0.$$

For the third-order system to be stable, it is necessary and sufficient that the coefficients be positive and $a_2 a_1 > a_0 a_3$. The condition when $a_2 a_1 = a_0 a_3$ results in a marginal stability case, and one pair of roots lies on the imaginary axis in the s-plane. This marginal case is recognized as Case 3 because there is a zero in the first column when $a_2 a_1 = a_0 a_3$. It will be discussed under Case 3.

As a final example of characteristic equations that result in no zero elements in the first row, let us consider the polynomial

$$q(s) = (s - 1 + j\sqrt{7})(s - 1 - j\sqrt{7})(s + 3) = s^3 + s^2 + 2s + 24. \tag{6.9}$$

The polynomial satisfies all the necessary conditions because all the coefficients exist and are positive. Therefore, utilizing the Routh array, we have

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 1 & 24 \\ s^1 & -22 & 0 \\ s^0 & 24 & 0 \end{array}$$

Because two changes in sign appear in the first column, we find that two roots of $q(s)$ lie in the right-hand plane, and our prior knowledge is confirmed. ■

Case 2. **There is a zero in the first column, but some other elements of the row containing the zero in the first column are nonzero.** If only one element in the array is zero, it may be replaced with a small positive number, ϵ, that is allowed to approach zero after completing the array. For example, consider the following characteristic polynomial:

$$q(s) = s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10. \tag{6.10}$$

The Routh array is then

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 11 \\ s^4 & 2 & 4 & 10 \\ s^3 & \epsilon & 6 & 0 \\ s^2 & c_1 & 10 & 0 \\ s^1 & d_1 & 0 & 0 \\ s^0 & 10 & 0 & 0 \end{array}$$

where

$$c_1 = \frac{4\epsilon - 12}{\epsilon} \quad \text{and} \quad d_1 = \frac{6c_1 - 10\epsilon}{c_1}.$$

When $0 < \epsilon \ll 1$, we find that $c_1 < 0$ and $d_1 > 0$. Therefore, there are two sign changes in the first column; hence the system is unstable with two roots in the right half-plane.

EXAMPLE 6.3 Unstable system

As a final example of the type of Case 2, consider the characteristic polynomial

$$q(s) = s^4 + s^3 + s^2 + s + K, \tag{6.11}$$

where we desire to determine the gain K that results in marginal stability. The Routh array is then

$$\begin{array}{c|ccc} s^4 & 1 & 1 & K \\ s^3 & 1 & 1 & 0 \\ s^2 & \epsilon & K & 0 \\ s^1 & c_1 & 0 & 0 \\ s^0 & K & 0 & 0 \end{array},$$

where

$$c_1 = \frac{\epsilon - K}{\epsilon}.$$

When $0 < \epsilon \ll 1$ and $K > 0$, we find that $c_1 < 0$. Therefore, there are two sign changes in the first column; hence, the system is unstable with two roots in the right half-plane. When $0 < \epsilon \ll 1$ and $K < 0$, we find that $c_1 > 0$, but because the last term in the first column is equal to K, we have a sign change in the first column; hence, the system is unstable with one root in the right half-plane. Consequently, the system is unstable for all values of gain K. ■

Case 3. **There is a zero in the first column, and the other elements of the row containing the zero are also zero.** Case 3 occurs when all the elements in one row are zero or when the row consists of a single element that is zero. This condition occurs when the polynomial contains singularities that are symmetrically located about the origin of the s-plane. Therefore, Case 3 occurs when factors such as $(s + \sigma)(s - \sigma)$ or $(s + j\omega)(s - j\omega)$ occur. This problem is circumvented by utilizing the **auxiliary polynomial**, $U(s)$, which immediately precedes the zero entry in the Routh array. The order of the auxiliary polynomial is always even and indicates the number of symmetrical root pairs.

To illustrate this approach, let us consider a third-order system with the characteristic polynomial

$$q(s) = s^3 + 2s^2 + 4s + K, \tag{6.12}$$

where K is an adjustable loop gain. The Routh array is then

$$\begin{array}{c|cc} s^3 & 1 & 4 \\ s^2 & 2 & K \\ s^1 & \dfrac{8-K}{2} & 0 \\ s^0 & K & 0 \end{array}.$$

For a stable system, we require that

$$0 < K < 8.$$

When $K = 8$, we have two roots on the $j\omega$-axis and a marginal stability case. Note that we obtain a row of zeros (Case 3) when $K = 8$. The auxiliary polynomial, $U(s)$, is the equation of the row preceding the row of zeros. The equation of the row preceding the row of zeros is, in this case, obtained from the s^2-row. We recall that this row contains the coefficients of the even powers of s, and therefore we have

$$U(s) = 2s^2 + Ks^0 = 2s^2 + 8 = 2(s^2 + 4) = 2(s + j2)(s - j2). \quad (6.13)$$

When $K = 8$, the factors of the characteristic polynomial are

$$q(s) = (s + 2)(s + j2)(s - j2). \quad (6.14)$$

Case 4. Repeated roots of the characteristic equation on the $j\omega$-axis. If the $j\omega$-axis roots of the characteristic equation are simple, the system is neither stable nor unstable; it is instead called marginally stable, since it has an undamped sinusoidal mode. If the $j\omega$-axis roots are repeated, the system response will be unstable with a form $t\sin(\omega t + \phi)$. The Routh–Hurwitz criteria will not reveal this form of instability [20].

Consider the system with a characteristic polynomial

$$q(s) = (s + 1)(s + j)(s - j)(s + j)(s - j) = s^5 + s^4 + 2s^3 + 2s^2 + s + 1.$$

The Routh array is

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 1 \\ s^4 & 1 & 2 & 1 \\ s^3 & \epsilon & \epsilon & 0 \\ s^2 & 1 & 1 & \\ s^1 & \epsilon & 0 & \\ s^0 & 1 & & \end{array}.$$

When $0 < \epsilon \ll 1$, we note the absence of sign changes in the first column. However, as $\epsilon \rightarrow 0$, we obtain a row of zero at the s^3 line and a now of zero at the s^1 line. The auxiliary polynomial at the s^2 line is $s^2 + 1$, and the auxiliary polynomial at the s^4 line is $s^4 + 2s^2 + 1 = (s^2 + 1)^2$, indicating the repeated roots on the $j\omega$-axis. Hence, the system is unstable.

EXAMPLE 6.4 Fifth-order system with roots on the $j\omega$-axis

Consider the characteristic polynomial

$$q(s) = s^5 + s^4 + 4s^3 + 24s^2 + 3s + 63. \tag{6.15}$$

The Routh array is

$$\begin{array}{c|ccc} s^5 & 1 & 4 & 3 \\ s^4 & 1 & 24 & 63 \\ s^3 & -20 & -60 & 0. \\ s^2 & 21 & 63 & 0 \\ s^1 & 0 & 0 & 0 \end{array}$$

Therefore, the auxiliary polynomial is

$$U(s) = 21s^2 + 63 = 21(s^2 + 3) = 21(s + j\sqrt{3})(s - j\sqrt{3}), \tag{6.16}$$

which indicates that two roots are on the imaginary axis. To examine the remaining roots, we divide by the auxiliary polynomial to obtain

$$\frac{q(s)}{s^2 + 3} = s^3 + s^2 + s + 21.$$

Establishing a Routh array for this equation, we have

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 1 & 21 \\ s^1 & -20 & 0^{\cdot} \\ s^0 & 21 & 0 \end{array}$$

The two changes in sign in the first column indicate the presence of two roots in the right-hand plane, and the system is unstable. The roots in the right-hand plane are $s = +1 \pm j\sqrt{6}$. ■

EXAMPLE 6.5 Welding control

Large welding robots are used in today's auto plants. The welding head is moved to different positions on the auto body, and a rapid, accurate response is required. A block diagram of a welding head positioning system is shown in Figure 6.5.

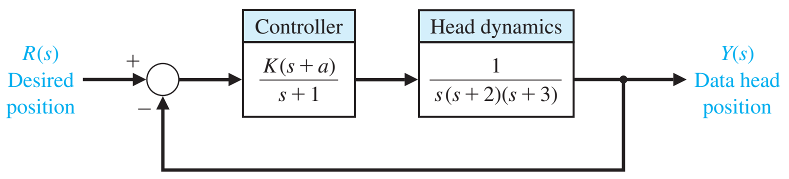


FIGURE 6.5
Welding head position control.

We desire to determine the range of K and a for which the system is stable. The characteristic equation is

$$1 + G(s) = 1 + \frac{K(s + a)}{s(s + 1)(s + 2)(s + 3)} = 0.$$

Therefore, $q(s) = s^4 + 6s^3 + 11s^2 + (K + 6)s + Ka = 0$. Establishing the Routh array, we have

$$\begin{array}{c|ccc} s^4 & 1 & 11 & Ka \\ s^3 & 6 & K + 6 & \\ s^2 & b_3 & Ka & \\ s^1 & c_3 & & \\ s^0 & Ka & & \end{array},$$

where

$$b_3 = \frac{60 - K}{6} \quad \text{and} \quad c_3 = \frac{b_3(K + 6) - 6Ka}{b_3}.$$

The coefficient c_3 sets the acceptable range of K and a, while b_3 requires that K be less than 60. Requiring $c_3 \geq 0$, we obtain

$$(K - 60)(K + 6) + 36Ka \leq 0.$$

The required relationship between K and a is then

$$a \leq \frac{(60 - K)(K + 6)}{36K}$$

when a is positive. Therefore, if $K = 40$, we require $a \leq 0.639$. ■

Suppose we write the characteristic equation of an nth-order system as

$$s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \cdots + a_1 s + \omega_n{}^n = 0.$$

We divide through by $\omega_n{}^n$ and use $\overset{*}{s} = s/\omega_n$ to obtain the normalized form of the characteristic equation:

$$\overset{*}{s}{}^n + b\overset{*}{s}{}^{n-1} + c\overset{*}{s}{}^{n-2} + \cdots + 1 = 0.$$

For example, we normalize

$$s^3 + 5s^2 + 2s + 8 = 0$$

by dividing through by $8 = \omega_n{}^3$, obtaining

$$\frac{s^3}{\omega_n{}^3} + \frac{5}{2}\frac{s^2}{\omega_n{}^2} + \frac{2}{4}\frac{s}{\omega_n} + 1 = 0,$$

or

$$\overset{*}{s}{}^3 + 2.5\overset{*}{s}{}^2 + 0.5\overset{*}{s} + 1 = 0,$$

Table 6.1 The Routh–Hurwitz Stability Criterion

n	Characteristic Equation	Criterion
2	$s^2 + bs + 1 = 0$	$b > 0$
3	$s^3 + bs^2 + cs + 1 = 0$	$bc - 1 > 0$
4	$s^4 + bs^3 + cs^2 + ds + 1 = 0$	$bcd - d^2 - b^2 > 0$
5	$s^5 + bs^4 + cs^3 + ds^2 + es + 1 = 0$	$bcd + b - d^2 - b^2e > 0$
6	$s^6 + bs^5 + cs^4 + ds^3 + es^2 + fs + 1 = 0$	$(bcd + bf - d^2 - b^2e)e + b^2c - bd - bc^2f - f^2 + bfe + cdf > 0$

Note: The equations are normalized by $(\omega_n)^n$.

where $\overset{*}{s} = s/\omega_n$. In this case, $b = 2.5$ and $c = 0.5$. Using this normalized form of the characteristic equation, we summarize the stability criterion for up to a sixth-order characteristic equation, as provided in Table 6.1. Note that $bc = 1.25$ and the system is stable.

6.3 THE RELATIVE STABILITY OF FEEDBACK CONTROL SYSTEMS

The verification of stability using the Routh–Hurwitz criterion provides only a partial answer to the question of stability. The Routh–Hurwitz criterion ascertains the absolute stability of a system by determining whether any of the roots of the characteristic equation lie in the right half of the s-plane. However, if the system satisfies the Routh–Hurwitz criterion and is stable, it is desirable to determine the **relative stability**; that is, it is interesting to investigate the relative damping of each root of the characteristic equation. The relative stability of a system can be defined as the property that is measured by the relative real part of each root or pair of roots. Thus, root r_2 is relatively more stable than the roots $r_1, \hat{r}_1$, as shown in Figure 6.6. The relative stability of a system can also be defined in terms of the relative damping coefficients ζ of each complex root pair and, therefore, in terms of the speed of response and overshoot instead of settling time.

The investigation of the relative stability of each root is important because the location of the closed-loop poles in the s-plane determines the performance of the system. Thus, we reexamine the characteristic polynomial $q(s)$ and consider several methods for the determination of relative stability.

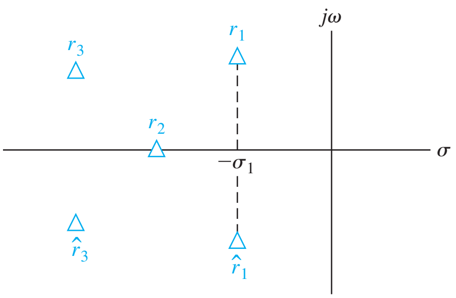


FIGURE 6.6 Root locations in the s-plane.

Because the relative stability of a system is determined by the location of the roots of the characteristic equation, a first approach using an s-plane formulation is to extend the Routh–Hurwitz criterion to ascertain relative stability. This can be accomplished by utilizing a change of variable, which shifts the s-plane axis in order to utilize the Routh–Hurwitz criterion. Examining Figure 6.6, we notice that a shift of the vertical axis in the s-plane to $-\sigma_1$ will result in the roots $r_1, \hat{r}_1$ appearing on the shifted axis. The correct magnitude to shift the vertical axis must be obtained on a trial-and-error basis. Then, without solving the fifth-order polynomial $q(s)$, we may determine the real part of the dominant roots $r_1, \hat{r}_1$.

EXAMPLE 6.6 Axis shift

Consider the third-order characteristic equation

$$q(s) = s^3 + 4s^2 + 6s + 4. \tag{6.17}$$

Setting the shifted variable s_n equal to $s + 1$, we obtain

$$(s_n - 1)^3 + 4(s_n - 1)^2 + 6(s_n - 1) + 4 = s_n^{\,3} + s_n^{\,2} + s_n + 1. \tag{6.18}$$

Then the Routh array is established as

$$\begin{array}{c|cc} s_n^{\,3} & 1 & 1 \\ s_n^{\,2} & 1 & 1 \\ s_n^{\,1} & 0 & 0 \\ s_n^{\,0} & 1 & 0 \end{array}.$$

There are roots on the shifted imaginary axis that can be obtained from the auxiliary polynomial

$$U(s_n) = s_n^{\,2} + 1 = (s_n + j)(s_n - j) = (s + 1 + j)(s + 1 - j). \tag{6.19}$$ ■

The shifting of the s-plane axis to ascertain the relative stability of a system is a very useful approach, particularly for higher-order systems with several pairs of closed-loop complex conjugate roots.

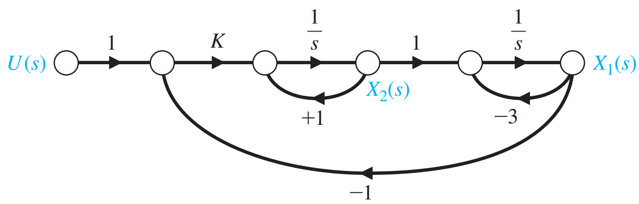
6.4 THE STABILITY OF STATE VARIABLE SYSTEMS

The stability of a system modeled by a state variable flow graph model can be readily ascertained. If the system we are investigating is represented by a signal-flow graph state model, we obtain the characteristic equation by evaluating the flow graph determinant. If the system is represented by a block diagram model we obtain the characteristic equation using the block diagram reduction methods.

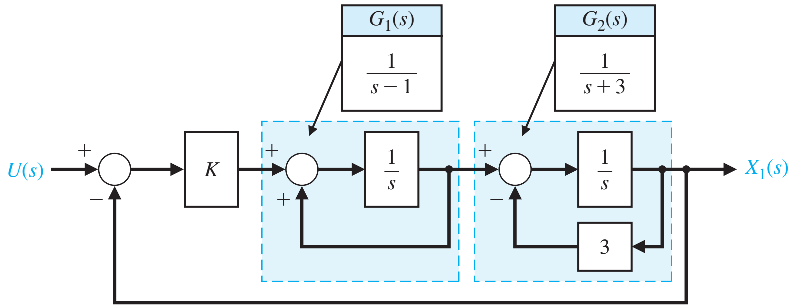
EXAMPLE 6.7 Stability of a second-order system

A second-order system is described by the two first-order differential equations

$$\dot{x}_1 = -3x_1 + x_2 \quad \text{and} \quad \dot{x}_2 = +1x_2 - Kx_1 + Ku, \tag{6.20}$$



(a)



(b)

FIGURE 6.7 (a) Flow graph model for state variable equations of Example 6.7. (b) Block diagram model.

where $u(t)$ is the input. The flow graph model of this set of differential equations is shown in Figure 6.7(a) and the block diagram model is shown in Figure 6.7(b).

Using Mason's signal-flow gain formula, we note three loops:

$$L_1 = s^{-1}, \qquad L_2 = -3s^{-1}, \quad \text{and} \quad L_3 = -Ks^{-2},$$

where L_1 and L_2 do not share a common node. Therefore, the determinant is

$$\Delta = 1 - (L_1 + L_2 + L_3) + L_1L_2 = 1 - (s^{-1} - 3s^{-1} - Ks^{-2}) + (-3s^{-2}).$$

We multiply by s^2 to obtain the characteristic equation

$$s^2 + 2s + (K - 3) = 0.$$

Since all coefficients must be positive, we require $K > 3$ for stability. A similar analysis can be undertaken using the block diagram. Closing the two feedback loops yields the two transfer functions

$$G_1(s) = \frac{1}{s - 1} \quad \text{and} \quad G_2(s) = \frac{1}{s + 3},$$

as illustrated in Figure 6.7(b). The closed loop transfer function is thus

$$T(s) = \frac{KG_1(s)G_2(s)}{1 + KG_1(s)G_2(s)}.$$

Therefore, the characteristic equation is

$$\Delta(s) = 1 + KG_1(s)G_2(s) = 0,$$

or

$$\Delta(s) = (s - 1)(s + 3) + K = s^2 + 2s + (K - 3) = 0. \tag{6.21}$$

This confirms the results obtained using signal-flow graph techniques. ■

A method of obtaining the characteristic equation directly from the vector differential equation is based on the fact that the solution to the unforced system is an exponential function. The vector differential equation without input signals is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \tag{6.22}$$

where $\mathbf{x}$ is the state vector. The solution is of exponential form, and we can define a constant λ such that the solution of the system for one state can be of the form $x_i(t) = k_i e^{\lambda_i t}$. The λ_i are called the characteristic roots or eigenvalues of the system, which are simply the roots of the characteristic equation. If we let $\mathbf{x} = \mathbf{k}e^{\lambda t}$ and substitute into Equation (6.22), we have

$$\lambda \mathbf{k} e^{\lambda t} = \mathbf{A}\mathbf{k}e^{\lambda t}, \tag{6.23}$$

or

$$\lambda \mathbf{x} = \mathbf{A}\mathbf{x}. \tag{6.24}$$

Equation (6.24) can be rewritten as

$$(\lambda \mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}, \tag{6.25}$$

where $\mathbf{I}$ equals the identity matrix and $\mathbf{0}$ equals the null matrix. This set of simultaneous equations has a nontrivial solution if and only if the determinant vanishes—that is, only if

$$\det(\lambda \mathbf{I} - \mathbf{A}) = 0. \tag{6.26}$$

The nth-order equation in λ resulting from the evaluation of this determinant is the characteristic equation, and the stability of the system can be readily ascertained.

EXAMPLE 6.8 Closed epidemic system

The vector differential equation of the epidemic system is given in Equation (3.63) and repeated here as

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -\alpha & -\beta & 0 \\ \beta & -\gamma & 0 \\ \alpha & \gamma & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

The characteristic equation is then

$$\begin{aligned} \det(\lambda \mathbf{I} - \mathbf{A}) &= \det\left\{\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} -\alpha & -\beta & 0 \\ \beta & -\gamma & 0 \\ \alpha & \gamma & 0 \end{bmatrix}\right\} \\ &= \det\begin{bmatrix} \lambda + \alpha & \beta & 0 \\ -\beta & \lambda + \gamma & 0 \\ -\alpha & -\gamma & \lambda \end{bmatrix} \\ &= \lambda[\lambda^2 + (\alpha + \gamma)\lambda + (\alpha\gamma + \beta^2)] = 0. \end{aligned}$$

Thus, we obtain the characteristic equation of the system. The additional root $\lambda = 0$ results from the definition of x_3 as the integral of $\alpha x_1 + \gamma x_2$, and x_3 does not affect the other state variables. Thus, the root $\lambda = 0$ indicates the integration connected with x_3. The characteristic equation indicates that the system is marginally stable when $\alpha + \gamma > 0$ and $\alpha\gamma + \beta^2 > 0$. ■

6.5 DESIGN EXAMPLES

In this section we present two illustrative examples. The first example is a tracked vehicle control problem. In this first example, stability issues are addressed employing the Routh–Hurwitz stability criterion and the outcome is the selection of two key system parameters. The second example illustrates the stability problem robot-controlled motorcycle and how Routh–Hurwitz can be used in the selection of controller gains during the design process. The robot-controlled motorcycle example highlights the design process with special attention to the impact of key controller parameters on stability.

EXAMPLE 6.9 **Tracked vehicle turning control**

The design of a turning control for a tracked vehicle involves the selection of two parameters [8]. In Figure 6.8, the system shown in part (a) has the model shown in part (b). The two tracks are operated at different speeds in order to turn the vehicle. We must select K and a so that the system is stable and the steady-state error for a ramp command is less than or equal to 24% of the magnitude of the command.

The characteristic equation of the feedback system is

$$1 + G_cG(s) = 0,$$

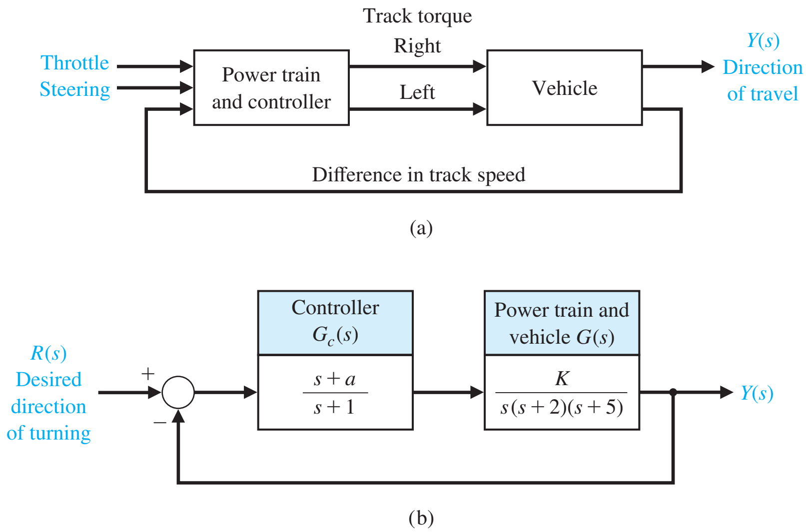


FIGURE 6.8
(a) Turning control system for a two-track vehicle.
(b) Block diagram.

or

$$1 + \frac{K(s + a)}{s(s + 1)(s + 2)(s + 5)} = 0. \tag{6.27}$$

Therefore, we have

$$s(s + 1)(s + 2)(s + 5) + K(s + a) = 0,$$

or

$$s^4 + 8s^3 + 17s^2 + (K + 10)s + Ka = 0. \tag{6.28}$$

To determine the stable region for K and a, we establish the Routh array as

$$\begin{array}{c|ccc} s^4 & 1 & 17 & Ka \\ s^3 & 8 & K + 10 & 0 \\ s^2 & b_3 & Ka & \\ s^1 & c_3 & & \\ s^0 & Ka & & \end{array},$$

where

$$b_3 = \frac{126 - K}{8} \quad \text{and} \quad c_3 = \frac{b_3(K + 10) - 8Ka}{b_3}.$$

For the elements of the first column to be positive, we require that Ka, b_3, and c_3 be positive. Therefore, we require that

$$K < 126,$$
$$Ka > 0, \quad \text{and}$$
$$(K + 10)(126 - K) - 64Ka > 0. \tag{6.29}$$

The region of stability for $K > 0$ is shown in Figure 6.9. The steady-state error to a ramp input $r(t) = At$, $t > 0$ is

$$e_{ss} = A/K_v,$$

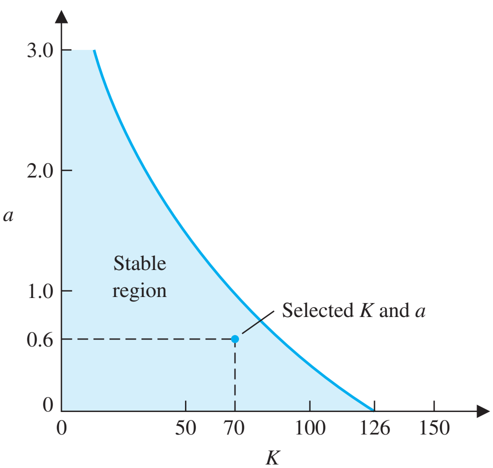


FIGURE 6.9
The stable region.

where

$$K_v = \lim_{s\to 0} sG_cG = Ka/10.$$

Therefore, we have

$$e_{ss} = \frac{10A}{Ka}. \tag{6.30}$$

When e_{ss} is equal to 23.8% of A, we require that $Ka = 42$. This can be satisfied by the selected point in the stable region when $K = 70$ and $a = 0.6$, as shown in Figure 6.9. Another acceptable design would be attained when $K = 50$ and $a = 0.84$. We can calculate a series of possible combinations of K and a that can satisfy $Ka = 42$ and that lie within the stable region, and all will be acceptable design solutions. However, not all selected values of K and a will lie within the stable region. Note that K cannot exceed 126. ■

EXAMPLE 6.10 Robot-controlled motorcycle

Consider the robot-controlled motorcycle shown in Figure 6.10. The motorcycle will move in a straight line at constant forward speed v. Let $\phi(t)$ denote the angle between the plane of symmetry of the motorcycle and the vertical. The desired angle $\phi_d(t)$ is equal to zero, thus

$$\phi_d(s) = 0.$$

The design elements highlighted in this example are illustrated in Figure 6.11. Using the Routh–Hurwitz stability criterion will allow us to get to the heart of the matter, that is, to develop a strategy for computing the controller gains while ensuring closed-loop stability. The control goal is

Control Goal

Control the motorcycle in the vertical position, and maintain the prescribed position in the presence of disturbances.

The variable to be controlled is

Variable to Be Controlled

The motorcycle position from vertical, $\phi(t)$.

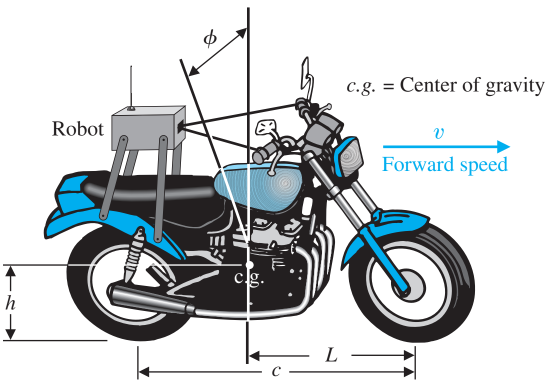


FIGURE 6.10 The robot-controlled motorcycle.

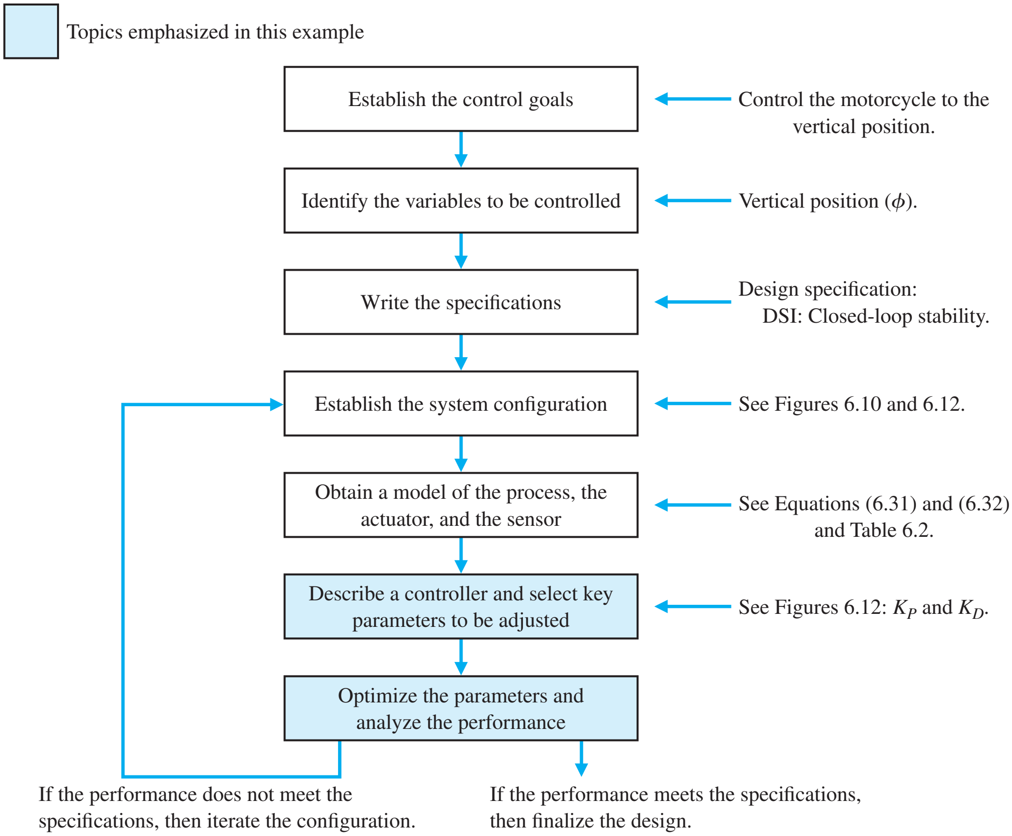


FIGURE 6.11 Elements of the control system design process emphasized in this robot-controlled motorcycle example.

Since our focus here is on stability rather than transient response characteristics, the control specifications will be related to stability only; transient performance is an issue that we need to address once we have investigated all the stability issues. The control design specification is

Design Specification

DS1 The closed-loop system must be stable.

The main components of the robot-controlled motorcycle are the motorcycle and robot, the controller, and the feedback measurements. The main subject of the chapter is not modeling, so we do not concentrate on developing the motorcycle dynamics model. We rely instead on the work of others (see [22]). The motorcycle model is given by

$$G(s) = \frac{1}{s^2 - \alpha_1}, \tag{6.31}$$

where $\alpha_1 = g/h$, $g = 9.806\ \text{m/s}^2$, and h is the height of the motorcycle center of gravity above the ground (see Figure 6.10). The motorcycle is unstable with poles at $s = \pm\sqrt{\alpha_1}$. The controller is given by

$$G_c(s) = \frac{\alpha_2 + \alpha_3 s}{\tau s + 1}, \tag{6.32}$$

where

$$\alpha_2 = v^2/(hc)$$

and

$$\alpha_3 = vL/(hc).$$

The forward speed of the motorcycle is denoted by v, and c denotes the wheelbase (the distance between the wheel centers). The length, L, is the horizontal distance between the front wheel axle and the motorcycle center of gravity. The time-constant of the controller is denoted by τ. This term represents the speed of response of the controller; smaller values of τ indicate an increased speed of response. Many simplifying assumptions are necessary to obtain the simple transfer function models in Equations (6.31) and (6.32).

Control is accomplished by turning the handlebar. The front wheel rotation about the vertical is not evident in the transfer functions. Also, the transfer functions assume a constant forward speed v which means that we must have another control system at work regulating the forward speed. Nominal motorcycle and robot controller parameters are given in Table 6.2.

Assembling the components of the feedback system gives us the system configuration shown in Figure 6.12. Examination of the configuration reveals that the robot controller block is a function of the physical system (h, c, and L), the operating conditions (v), and the robot time-constant (τ). No parameters need adjustment unless we physically change the motorcycle parameters and/or speed. In fact, in this example the parameters we want to adjust are in the feedback loop:

Select Key Tuning Parameters

Feedback gains K_P and K_D.

The key tuning parameters are not always in the forward path; in fact they may exist in any subsystem in the block diagram.

We want to use the Routh–Hurwitz technique to analyze the closed-loop system stability. What values of K_P and K_D lead to closed-loop stability? A related question that we can pose is, given specific values of K_P and K_D for the nominal system (that is, nominal values of α_1, α_2, α_3, and τ), how can the parameters themselves vary while still retaining closed-loop stability?

Table 6.2 Physical Parameters

τ	0.2 s
α_1	9 1/s^2
α_2	2.7 1/s^2
α_3	1.35 1/s
h	1.09 m
V	2.0 m/s
L	1.0 m
c	1.36 m

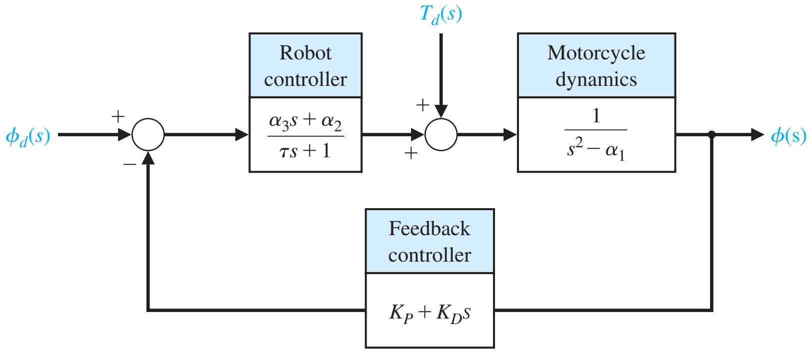


FIGURE 6.12 The robot-controlled motorcycle feedback system block diagram.

The closed-loop transfer function from $\phi_d(s)$ to $\phi(s)$ is

$$T(s) = \frac{\alpha_2 + \alpha_3 s}{\Delta(s)},$$

where

$$\Delta(s) = \tau s^3 + (1 + K_D\alpha_3)s^2 + (K_D\alpha_2 + K_P\alpha_3 - \tau\alpha_1)s + K_P\alpha_2 - \alpha_1.$$

The question that we need to answer is for what values of K_P and K_D does the characteristic equation $\Delta(s) = 0$ have all roots in the left half-plane?

We can set up the following Routh array:

$$\begin{array}{c|cc} s^3 & \tau & K_D\alpha_2 + K_P\alpha_3 - \tau\alpha_1 \\ s^2 & 1 + K_D\alpha_3 & K_P\alpha_2 - \alpha_1 \\ s & a & \\ 1 & K_P\alpha_2 - \alpha_1 & \end{array}$$

where

$$a = \frac{(1 + K_D\alpha_3)(K_D\alpha_2 + K_P\alpha_3 - \tau\alpha_1) - \tau(\alpha_2 K_P - \alpha_1)}{1 + K_D\alpha_3}.$$

By inspecting column 1, we determine that for stability we require

$$\tau > 0,\ K_D > -1/\alpha_3,\ K_P > \alpha_1/\alpha_2, \text{ and } a > 0.$$

Choosing $K_D > 0$ satisfies the second inequality (note that $\alpha_3 > 0$). In the event $\tau = 0$, we would reformulate the characteristic equation and rework the Routh array. We need to determine the conditions on K_P and K_D such that $a > 0$. We find that $a > 0$ implies that the following relationship must be satisfied:

$$\alpha_2\alpha_3 K_D^2 + (\alpha_2 - \tau\alpha_1\alpha_3 + \alpha_3^2 K_P)K_D + (\alpha_3 - \tau\alpha_2)K_P > 0. \quad (6.33)$$

Using the nominal values of the parameters α_1, α_2, α_3, and τ (see Table 6.2), for all $K_D > 0$ and $K_P > 3.33$, the left hand-side of Equation (6.33) is positive, hence $a > 0$. Taking into account all the inequalities, a valid region for selecting the gains is $K_D > 0$ and $K_P > \alpha_1/\alpha_2 = 3.33$.

Selecting any point (K_P, K_D) in the stability region yields a valid (that is, stable) set of gains for the feedback loop. For example, selecting

$$K_P = 10 \text{ and } K_D = 5$$

yields a stable closed-loop system. The closed-loop poles are

$$s_1 = -35.2477, s_2 = -2.4674, \text{ and } s_3 = -1.0348.$$

Since all the poles have negative real parts, we know the system response to any bounded input will be bounded.

For this robot-controlled motorcycle, we do not expect to have to respond to nonzero command inputs (that is, $\phi_d(t) \neq 0$) since we want the motorcycle to remain upright, and we certainly want to remain upright in the presence of external disturbances. The transfer function for the disturbance $T_d(s)$ to the output $\phi(s)$ without feedback is

$$\phi(s) = \frac{1}{s^2 - \alpha_1} T_d(s).$$

The characteristic equation is

$$q(s) = s^2 - \alpha_1 = 0.$$

The system poles are

$$s_1 = -\sqrt{\alpha_1} \text{ and } s_2 = +\sqrt{\alpha_1}.$$

Thus we see that the motorcycle is unstable; it possesses a pole in the right half-plane. Without feedback control, any external disturbance will result in the motorcycle falling over. Clearly the need for a control system (usually provided by the human rider) is necessary. With the feedback and robot controller in the loop, the closed-loop transfer function from the disturbance to the output is

$$\frac{\phi(s)}{T_d(s)} = \frac{\tau s + 1}{\tau s^3 + (1 + K_D\alpha_3)s^2 + (K_D\alpha_2 + K_P\alpha_3 - \tau\alpha_1)s + K_P\alpha_2 - \alpha_1}.$$

The response to a step disturbance is shown in Figure 6.13; the response is stable. The control system manages to keep the motorcycle upright, although it is tilted at about $\phi = 0.055$ rad $= 3.18$ deg.

It is important to give the robot the ability to control the motorcycle over a wide range of forward speeds. Is it possible for the robot, with the feedback gains as selected ($K_P = 10$ and $K_D = 5$), to control the motorcycle as the velocity varies? From experience we know that at slower speeds a bicycle becomes more difficult to control. We expect to see the same characteristics in the stability analysis of our system. Whenever possible, we try to relate the engineering problem at hand to real-life experiences. This helps to develop intuition that can be used as a reasonableness check on our solution.

A plot of the roots of the characteristic equation as the forward speed v varies is shown in Figure 6.14. The data in the plot were generated using the nominal values of the feedback gains, $K_P = 10$ and $K_D = 5$. We selected these gains for the case where $v = 2$ m/s. Figure 6.14 shows that as v increases, the roots of the

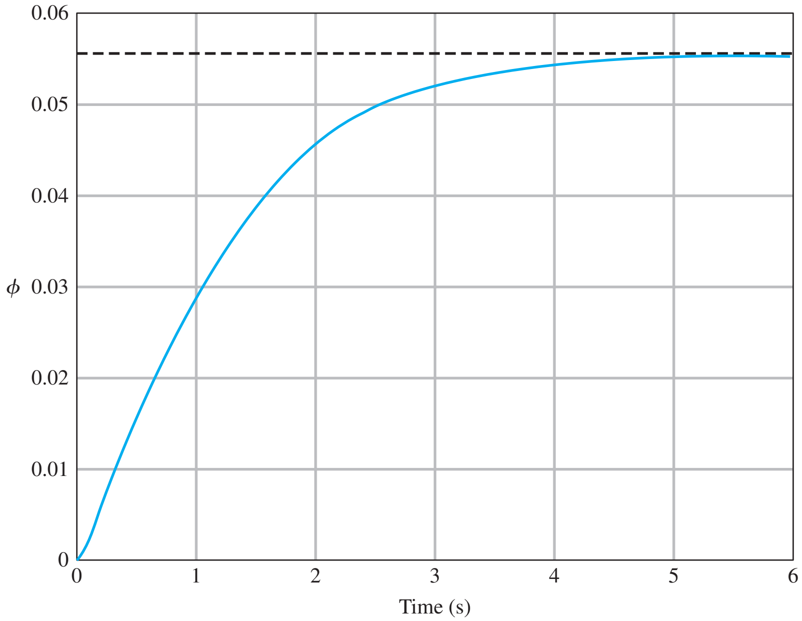


FIGURE 6.13 Disturbance response with $K_P = 10$ and $K_D = 5$.

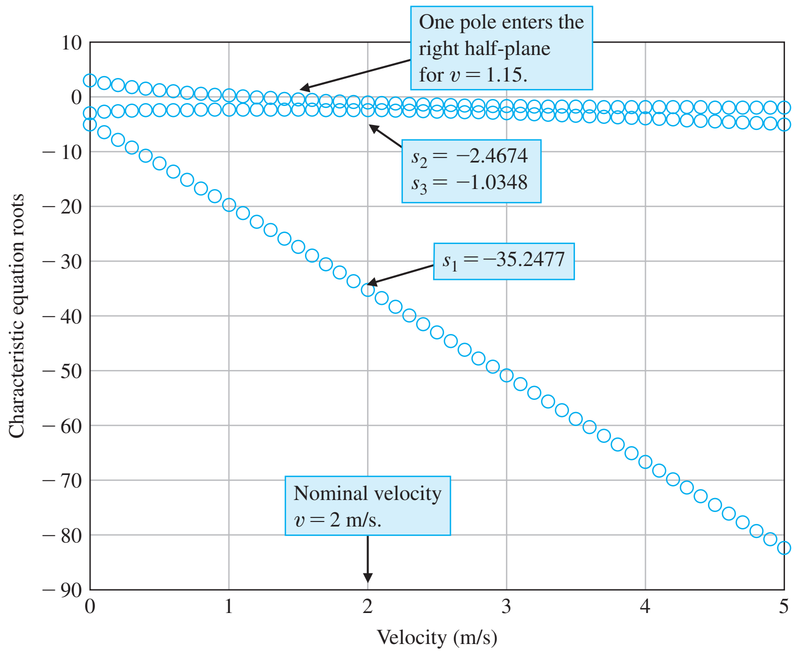


FIGURE 6.14 Roots of the characteristic equation as the motorcycle velocity varies.

characteristic equation remain stable (that is, in the left half-plane) with all points negative. But as the motorcycle forward speed decreases, the roots move toward zero, with one root becoming positive at $v = 1.15$ m/s. At the point where one root is positive, the motorcycle is unstable. ■

6.6 SYSTEM STABILITY USING CONTROL DESIGN SOFTWARE

In this section we will see how the computer can assist us in the stability analysis by providing an easy and accurate method for computing the poles of the characteristic equation. For the case of the characteristic equation as a function of a single parameter, it will be possible to generate a plot displaying the movement of the poles as the parameter varies. The section concludes with an example.

The function introduced in this section is the function for, which is used to repeat a number of statements a specific number of times.

Routh–Hurwitz Stability. As stated earlier, the Routh–Hurwitz criterion is a necessary and sufficient criterion for stability. Given a characteristic equation with fixed coefficients, we can use Routh–Hurwitz to determine the number of roots in the right half-plane. For example, consider the characteristic equation

$$q(s) = s^3 + s^2 + 2s + 24 = 0$$

associated with the closed-loop control system shown in Figure 6.15. The corresponding Routh–Hurwitz array is shown in Figure 6.16. The two sign changes in the first column indicate that there are two roots of the characteristic polynomial in the right half-plane; hence, the closed-loop system is unstable. We can verify the Routh–Hurwitz result by directly computing the roots of the characteristic equation, as shown in Figure 6.17, using the pole function. Recall that the pole function computes the system poles.

Whenever the characteristic equation is a function of a single parameter, the Routh–Hurwitz method can be utilized to determine the range of values that the parameter may take while maintaining stability. Consider the closed-loop feedback system in Figure 6.18. The characteristic equation is

$$q(s) = s^3 + 2s^2 + 4s + K = 0.$$

Using a Routh–Hurwitz approach, we find that we require $0 < K < 8$ for stability (see Equation 6.12). We can verify this result graphically. As shown in Figure 6.19(b),

FIGURE 6.15 Closed-loop control system with $T(s) = Y(s)/R(s) = 1/(s^3 + s^2 + 2s + 24)$.

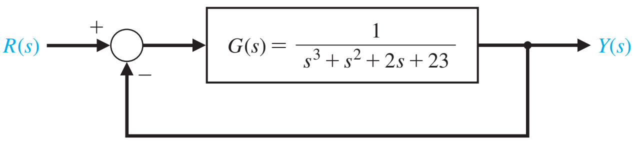
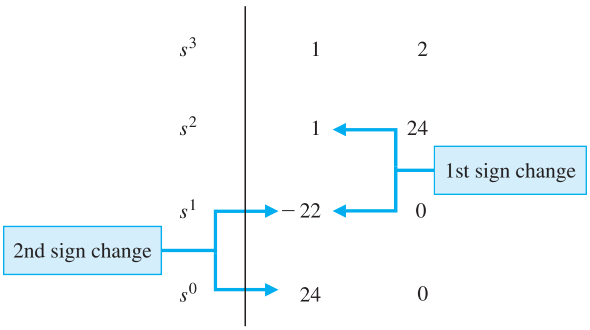


FIGURE 6.16 Routh array for the closed-loop control system with $T(s) = Y(s)/R(s) = 1/(s^3 + s^2 + 2s + 24)$.



```
>>numg=[1]; deng=[1 1 2 23]; sysg=tf(numg,deng);
>>sys=feedback(sysg,[1]);
>>pole(sys)

ans =

   -3.0000
    1.0000 + 2.6458i
    1.0000 - 2.6458i
```

Unstable poles

FIGURE 6.17 Using the **pole** function to compute the closed-loop control system poles of the system shown in Figure 6.16.

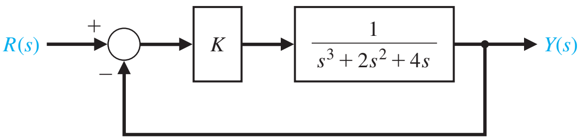


FIGURE 6.18 Closed-loop control system with $T(s) = Y(s)/R(s) = K/(s^3 + 2s^2 + 4s + 4)$.

we establish a vector of values for K at which we wish to compute the roots of the characteristic equation. Then using the roots function, we calculate and plot the roots of the characteristic equation, as shown in Figure 6.19(a). It can be seen that as K increases, the roots of the characteristic equation move toward the right half-plane as the gain tends toward $K = 8$, and eventually into the right half-plane when $K > 8$.

The script in Figure 6.19 contains the for function. This function provides a mechanism for repeatedly executing a series of statements a given number of times. The for function connected to an end statement sets up a repeating calculation loop. Figure 6.20 describes the for function format and provides an illustrative example of its usefulness. The example sets up a loop that repeats ten times. During the ith iteration, where $1 \leq i \leq 10$, the ith element of the vector **a** is set equal to 20, and the scalar b is recomputed.

The Routh–Hurwitz method allows us to make definitive statements regarding absolute stability of a linear system. The method does not address the issue of relative stability, which is directly related to the location of the roots of the characteristic equation. Routh–Hurwitz tells us how many poles lie in the right half-plane, but not the specific location of the poles. With control design software, we can easily calculate the poles explicitly, thus allowing us to comment on the relative stability.

EXAMPLE 6.11 **Tracked vehicle control**

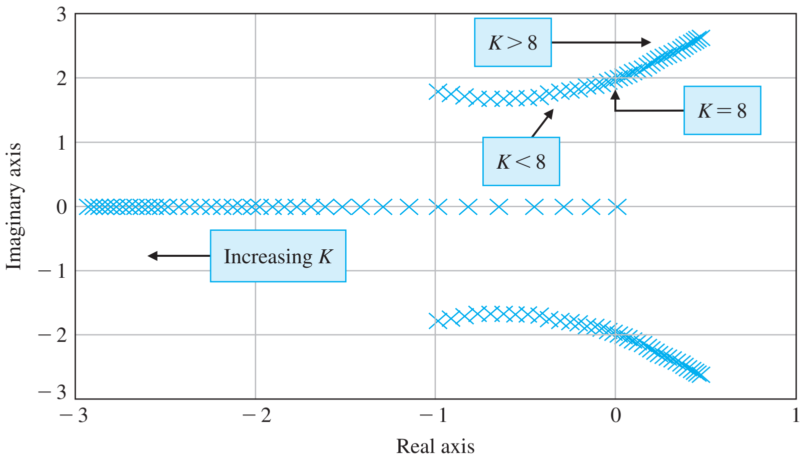
The block diagram of the control system for the two-track vehicle is shown in Figure 6.8. The design objective is to find a and K such that the system is stable and the steady-state error for a ramp input is less than or equal to 24% of the command.

We can use the Routh–Hurwitz method to aid in the search for appropriate values of a and K. The closed-loop characteristic equation is

$$q(s) = s^4 + 8s^3 + 17s^2 + (K + 10)s + aK = 0.$$

Using the Routh array, we find that, for stability, we require that

$$K < 126, \quad \frac{126 - K}{8}(K + 10) - 8aK > 0, \quad \text{and} \quad aK > 0.$$



(a)

```
% This script computes the roots of the characteristic
% equation q(s) = s^3 + 2 s^2 + 4 s + K for 0<K<20
%
K=[0:0.5:20];
for i=1:length(K)
   q=[1 2 4 K(i)];
   p(:,i)=roots(q);
end
plot(real(p),imag(p),'x'), grid
xlabel('Real axis'), ylabel('Imaginary axis')
```

Loop for roots as a function of K

(b)

FIGURE 6.19 (a) Plot of root locations of $q(s) = s^3 + 2s^2 + 4s + K$ for $0 \leq K \leq 20$. (b) m-file script.

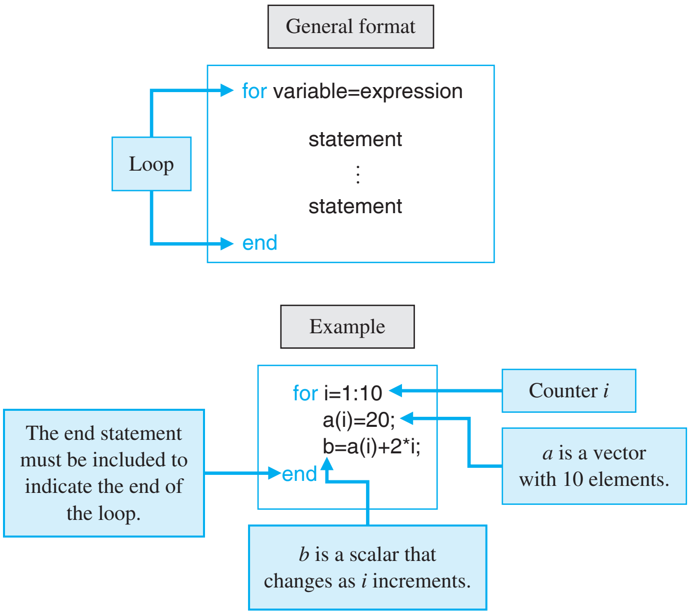
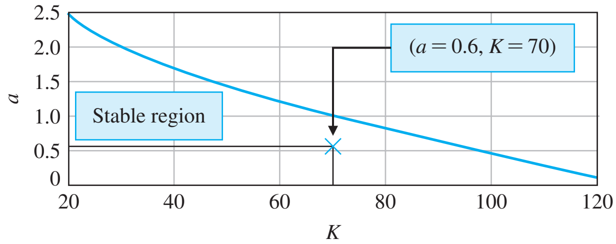


FIGURE 6.20 The **for** function and an illustrative example.



(a)

```
% The a-K stability region for the two track vehicle
% control problem
%
a=[0.1:0.01:3.0]; K=[20:1:120];
x=0*K; y=0*K;
n=length(K); m=length(a);
for i=1:n
 for j=1:m
  q=[1, 8, 17, K(i)+10, K(i)*a(j)];
  p=roots(q);
  if max(real(p)) > 0, x(i)=K(i); y(i)=a(j-1); break; end
 end
end
plot(x,y), grid, xlabel('K'), ylabel('a')
```

Range of a and K.

Initialize plot vectors as zero vectors of appropriate lengths.

Characteristic polynomial.

For a given value of K, determine first value of a for instability.

(b)

FIGURE 6.21 (a) Stability region for a and K for two-track vehicle turning control. (b) m-file script.

For positive K, it follows that we can restrict our search to $0 < K < 126$ and $a > 0$. Our approach will be to use the computer to help find a parameterized a versus K region in which stability is assured. Then we can find a set of (a, K) belonging to the stable region such that the steady-state error specification is met. This procedure, shown in Figure 6.21, involves selecting a range of values for a and K and computing the roots of the characteristic polynomial for specific values of a and K. For each value of K, we find the first value of a that results in at least one root of the characteristic equation in the right half-plane. The process is repeated until the entire selected range of a and K is exhausted. The plot of the (a, K) pairs defines the separation between the stable and unstable regions. The region to the left of the plot of a versus K in Figure 6.21 is the stable region.

If we assume that $r(t) = At$, $t > 0$, then the steady-state error is

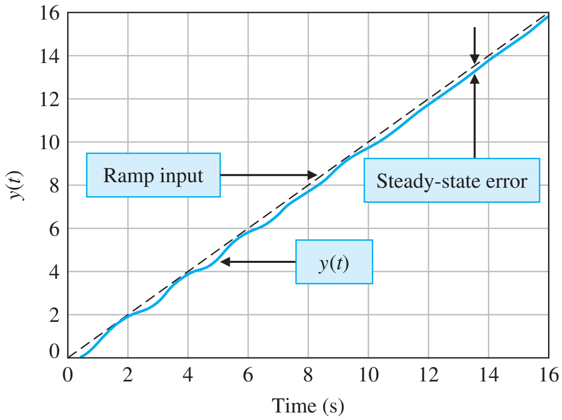
$$e_{ss} = \lim_{s \to 0} s \cdot \frac{s(s+1)(s+2)(s+5)}{s(s+1)(s+2)(s+5) + K(s+a)} \frac{A}{s^2} = \frac{10A}{aK},$$

where we have used the fact that

$$E(s) = \frac{1}{1 + G_c G(s)} R(s) = \frac{s(s+1)(s+2)(s+5)}{s(s+1)(s+2)(s+5) + K(s+a)} R(s).$$

Given the steady-state specification, $e_{ss} < 0.24A$, we find that the specification is satisfied when

$$\frac{10A}{aK} < 0.24A,$$



(a)

```
% Two-track vehicle turning control ramp response
% with a=0.6 and K=70.
%
t=[0:0.01:16]; u=t;
numgc=[1 0.6]; dengc=[1 1]; sysgc=tf(numgc,dengc);
numg=[70]; deng=[1 7 10 0]; sysg=tf(numg,deng);
sysa=series(sysgc,sysg);
sys=feedback(sysa,[1]);
y=lsim(sys,u,t);
plot(t,y,t,u,'--'), grid
xlabel('Time (s)'), ylabel('y(t)')
```

u = unit ramp input

Linear simulation

$a = 0.6$ and $K = 70$

(b)

FIGURE 6.22 (a) Ramp response for $a = 0.6$ and $K = 70$ for two-track vehicle turning control. (b) m-file script.

or

$$aK > 41.67. \tag{6.34}$$

Any values of a and K that lie in the stable region in Figure 6.21 and satisfy Equation (6.34) will lead to an acceptable design. For example, $K = 70$ and $a = 0.6$ will satisfy all the design requirements. The closed-loop transfer function (with $a = 0.6$ and $K = 70$) is

$$T(s) = \frac{70s + 42}{s^4 + 8s^3 + 17s^2 + 80s + 42}.$$

The associated closed-loop poles are

$$s = -7.0767,$$
$$s = -0.5781,$$
$$s = -0.1726 + 3.1995i, \quad \text{and}$$
$$s = -0.1726 - 3.1995i.$$

The corresponding unit ramp input response is shown in Figure 6.22. The steady-state error is less than 0.24, as desired. ■

The Stability of State Variable Systems. Now let us turn to determining the stability of systems described in state variable form. Suppose we have a system in state-space form as in Equation (6.22). The stability of the system can be evaluated with the characteristic equation associated with the system matrix **A**. The characteristic equation is

$$\det(s\mathbf{I} - \mathbf{A}) = 0. \tag{6.35}$$

The left-hand side of the characteristic equation is a polynomial in s. If all of the roots of the characteristic equation have negative real parts (i.e., $\mathrm{Re}(s_i) < 0$), then the system is stable.

When the system model is given in state variable form, we must calculate the characteristic polynomial associated with the **A** matrix. In this regard, we have several options. We can calculate the characteristic equation directly from Equation (6.35) by manually computing the determinant of $s\mathbf{I} - \mathbf{A}$. Then, we can compute the roots using the roots function to check for stability, or alternatively, we can use the Routh–Hurwitz method to detect any unstable roots. Unfortunately, the manual computations can become lengthy, especially if the dimension of **A** is large. We would like to avoid this manual computation if possible. As it turns out, the computer can assist in this endeavor.

The poly function can be used to compute the characteristic equation associated with **A**. The poly is used to form a polynomial from a vector of roots. It can also be used to compute the characteristic equation of **A**, as illustrated in Figure 6.23. The input matrix **A** is

$$\mathbf{A} = \begin{bmatrix} -8 & -16 & -6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

and the associated characteristic polynomial is

$$s^3 + 8s^2 + 16s + 6 = 0.$$

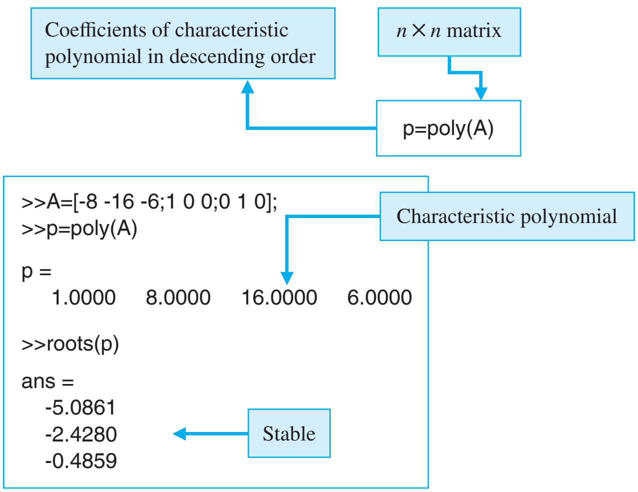


FIGURE 6.23 Computing the characteristic polynomial of **A** with the **poly** function.

If $\mathbf{A}$ is an $n \times n$ matrix, poly($\mathbf{A}$) is an $n + 1$ element row vector whose elements are the coefficients of the characteristic equation $\det(s\mathbf{I} - \mathbf{A}) = 0$.

6.7 SEQUENTIAL DESIGN EXAMPLE: DISK DRIVE READ SYSTEM

In this section, we will examine the stability of the disk drive read system as K_a is adjusted and then reconfigure the system.

Let us consider the system as shown in Figure 6.24. Initially, we consider the case where the switch is open. Then the closed-loop transfer function is

$$\frac{Y(s)}{R(s)} = \frac{K_a G_1(s) G_2(s)}{1 + K_a G_1(s) G_2(s)}, \tag{6.36}$$

where

$$G_1(s) = \frac{5000}{s + 1000} \quad \text{and} \quad G_2(s) = \frac{1}{s(s + 20)}.$$

The characteristic equation is

$$s^3 + 1020s^2 + 20000s + 5000K_a = 0. \tag{6.37}$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 20000 \\ s^2 & 1020 & 5000K_a \\ s^1 & b_1 & \\ s^0 & 5000K_a & \end{array},$$

where

$$b_1 = \frac{(20000)1020 - 5000K_a}{1020}.$$

The case $b_1 = 0$ results in marginal stability when $K_a = 4080$. Using the auxiliary equation, we have

$$s^2 + 20000 = 0,$$

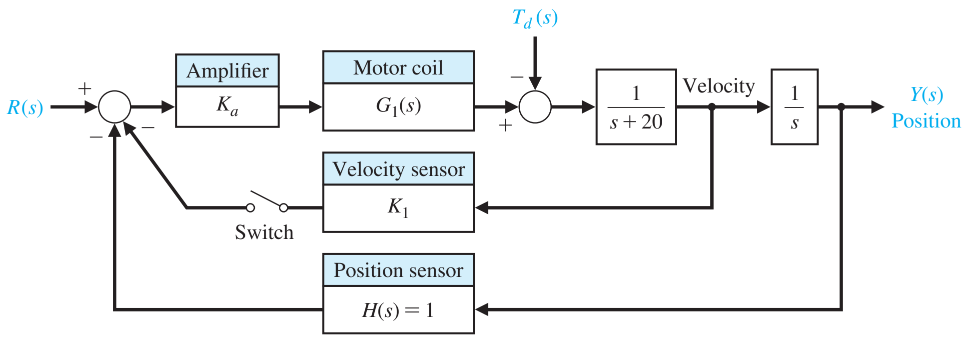


FIGURE 6.24 The closed-loop disk drive head system with an optional velocity feedback.

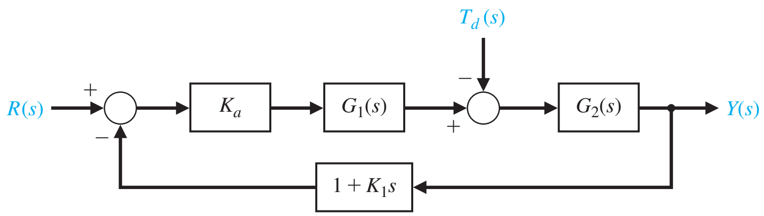


FIGURE 6.25 Equivalent system with the velocity feedback switch closed.

or the roots of the $j\omega$-axis are $s = \pm j141.4$. In order for the system to be stable, $K_a < 4080$.

Now let us add the velocity feedback by closing the switch in the system of Figure 6.24. The closed-loop transfer function for the system is then

$$\frac{Y(s)}{R(s)} = \frac{K_a G_1(s) G_2(s)}{1 + [K_a G_1(s) G_2(s)](1 + K_1 s)}, \tag{6.38}$$

since the feedback factor is equal to $1 + K_1 s$, as shown in Figure 6.25.

The characteristic equation is

$$1 + [K_a G_1(s) G_2(s)](1 + K_1 s) = 0,$$

or

$$s(s + 20)(s + 1000) + 5000K_a(1 + K_1 s) = 0.$$

Therefore, we have

$$s^3 + 1020s^2 + [20000 + 5000K_a K_1]s + 5000K_a = 0.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 20000 + 5000K_a K_1 \\ s^2 & 1020 & 5000K_a \\ s^1 & b_1 & \\ s^0 & 5000K_a & \end{array},$$

where

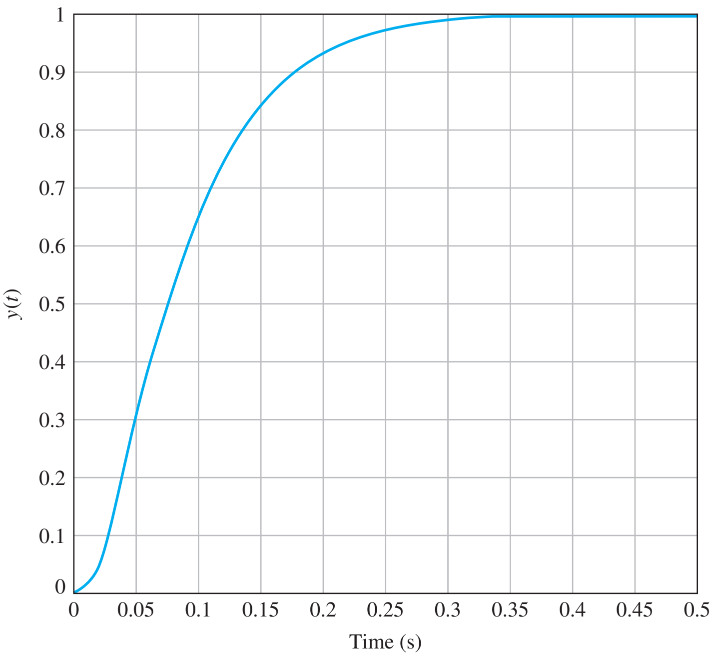
$$b_1 = \frac{1020(20000 + 5000K_a K_1) - 5000K_a}{1020}.$$

To guarantee stability, it is necessary to select the pair (K_a, K_1) such that $b_1 > 0$, where $K_a > 0$. When $K_1 = 0.05$ and $K_a = 100$, we can determine the system response using the script shown in Figure 6.26. The settling time (with a 2% criterion) is approximately $T_s = 260$ ms, and the percent overshoot is $P.O. = 0\%$. The system performance is summarized in Table 6.3. The performance specifications are nearly satisfied, and some iteration of K_1 is necessary to obtain the desired $T_s = 250$ ms.

```
Ka=100; K1=0.05;
ng1=[5000]; dg1=[1 1000]; sys1=tf(ng1,dg1);
ng2=[1]; dg2=[1 20 0]; sys2=tf(ng2,dg2);
nc=[K1 1]; dc=[0 1]; sysc=tf(nc,dc);
syso=series(Ka*sys1,sys2);
sys=feedback(syso,sysc); sys=minreal(sys);
t=[0:0.001:0.5];
y=step(sys,t); plot(t,y)
ylabel('y(t)'),xlabel('Time (s)'),grid
```

Select the velocity feedback gain K_1 and amplifier gain K_a.

(a)



(b)

FIGURE 6.26 Response of the system with velocity feedback. (a) m-file script. (b) Response with $K_a = 100$ and $K_1 = 0.05$.

Table 6.3 Performance of the Disk Drive System Compared to the Specifications

Performance Measure	Desired Value	Actual Response
Percent overshoot	Less than 5%	0%
Settling time	Less than 250 ms	260 ms
Maximum response to a unit disturbance	Less than 5×10^{-3}	2×10^{-3}

6.8 SUMMARY

In this chapter, we have considered the concept of the stability of a feedback control system. A definition of a stable system in terms of a bounded system response was outlined and related to the location of the poles of the system transfer function in the s-plane.

The Routh–Hurwitz stability criterion was introduced, and several examples were considered. The relative stability of a feedback control system was also considered in terms of the location of the poles and zeros of the system transfer function in the s-plane. The stability of state variable systems was considered.

SKILLS CHECK

In this section, we provide three sets of problems to test your knowledge: True or False, Multiple Choice, and Word Match. To obtain direct feedback, check your answers with the answer key provided at the conclusion of the end-of-chapter problems. Use the block diagram in Figure 6.27 as specified in the various problem statements.

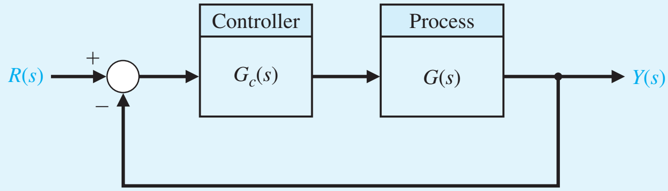


FIGURE 6.27 Block diagram for the Skills Check.

In the following **True or False** and **Multiple Choice** problems, circle the correct answer.

1. A stable system is a dynamic system with a bounded output response for any input. *True or False*

2. A marginally stable system has poles on the $j\omega$-axis. *True or False*

3. A system is stable if all poles lie in the right half-plane. *True or False*

4. The Routh–Hurwitz criterion is a necessary and sufficient criterion for determining the stability of linear systems. *True or False*

5. Relative stability characterizes the degree of stability. *True or False*

6. A system has the characteristic equation

$$q(s) = s^3 + 4Ks^2 + (5 + K)s + 10 = 0.$$

The range of K for a stable system is:

a. $K > 0.46$

b. $K < 0.46$

c. $0 < K < 0.46$

d. Unstable for all K

7. Utilizing the Routh–Hurwitz criterion, determine whether the following polynomials are stable or unstable:

$$p_1(s) = s^2 + 10s + 5 = 0,$$

$$p_2(s) = s^4 + s^3 + 5s^2 + 20s + 10 = 0.$$

a. $p_1(s)$ is stable, $p_2(s)$ is stable

b. $p_1(s)$ is unstable, $p_2(s)$ is stable

c. $p_1(s)$ is stable, $p_2(s)$ is unstable

d. $p_1(s)$ is unstable, $p_2(s)$ is unstable

8. Consider the feedback control system block diagram in Figure 6.27. Investigate closed-loop stability for $G_c(s) = K(s + 1)$ and $G(s) = \dfrac{1}{(s + 2)(s - 1)}$, for the two cases where $K = 1$ and $K = 3$.

   a. Unstable for $K = 1$ and stable for $K = 3$
   b. Unstable for $K = 1$ and unstable for $K = 3$
   c. Stable for $K = 1$ and unstable for $K = 3$
   d. Stable for $K = 1$ and stable for $K = 3$

9. Consider a unity negative feedback system in Figure 6.27 with loop transfer function where

$$L(s) = G_c(s)G(s) = \frac{K}{(1 + 0.5s)(1 + 0.5s + 0.25s^2)}.$$

   Determine the value of K for which the closed-loop system is marginally stable.

   a. $K = 10$
   b. $K = 3$
   c. The system is unstable for all K
   d. The system is stable for all K

10. A system is represented by $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -K & 10 \end{bmatrix}.$$

   The values of K for a stable system are

   a. $K < 1/2$
   b. $K > 1/2$
   c. $K = 1/2$
   d. The system is stable for all K

11. Use the Routh array to assist in computing the roots of the polynomial

$$q(s) = 2s^3 + 2s^2 + s + 1 = 0.$$

   a. $s_1 = -1; s_{2,3} = \pm\dfrac{\sqrt{2}}{2}j$
   b. $s_1 = 1; s_{2,3} = \pm\dfrac{\sqrt{2}}{2}j$
   c. $s_1 = -1; s_{2,3} = 1 \pm \dfrac{\sqrt{2}}{2}j$
   d. $s_1 = -1; s_{2,3} = 1$

12. Consider the following unity feedback control system in Figure 6.27 where

$$G(s) = \frac{1}{(s - 2)(s^2 + 10s + 45)} \text{ and } G_c(s) = \frac{K(s + 0.3)}{s}.$$

   The range of K for stability is

   a. $K < 260.68$
   b. $50.06 < K < 123.98$
   c. $100.12 < K < 260.68$
   d. The system is unstable for all $K > 0$

In Problems 13 and 14, consider the system represented in a state-space form

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -5 & -10 & -5 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 20 \end{bmatrix}u$$

$$y = [1 \quad 0 \quad 1]\mathbf{x}.$$

13. The characteristic equation is:

a. $q(s) = s^3 + 5s^2 - 10s - 6$

b. $q(s) = s^3 + 5s^2 + 10s + 5$

c. $q(s) = s^3 - 5s^2 + 10s - 5$

d. $q(s) = s^2 - 5s + 10$

14. Using the Routh–Hurwitz criterion, determine whether the system is stable, unstable, or marginally stable.

a. Stable

b. Unstable

c. Marginally stable

d. None of the above

15. A system has the block diagram representation as shown in Figure 6.27, where $G(s) = \dfrac{10}{(s+15)^2}$ and $G_c(s) = \dfrac{K}{s+80}$, where K is always positive. The limiting gain for a stable system is:

a. $0 < K < 28875$

b. $0 < K < 27075$

c. $0 < K < 25050$

d. Stable for all $K > 0$

In the following **Word Match** problems, match the term with the definition by writing the correct letter in the space provided.

a. Routh–Hurwitz criterion	A performance measure of a system.	_____
b. Auxiliary polynomial	A dynamic system with a bounded system response to a bounded input.	_____
c. Marginally stable	The property that is measured by the relative real part of each root or pair of roots of the characteristic equation.	_____
d. Stable system	A criterion for determining the stability of a system by examining the characteristic equation of the transfer function.	_____
e. Stability	The equation that immediately precedes the zero entry in the Routh array.	_____
f. Relative stability	A system description that reveals whether a system is stable or not stable without consideration of other system attributes such as degree of stability.	_____
g. Absolute stability	A system possesses this type of stability if the zero input response remains bounded as $t \rightarrow \infty$.	_____

EXERCISES

E6.1 A system has a characteristic equation $s^3 + Ks^2 + (1 + K)s + 6 = 0$. Determine the range of K for a stable system.

Answer: $K > 2$

E6.2 A system has a characteristic equation $s^3 + 15s^2 + 2s + 40 = 0$. Using the Routh–Hurwitz criterion, show that the system is unstable.

E6.3 A system has the characteristic equation $s^4 + 10s^3 + 32s^2 + 37s + 10 = 0$. Using the Routh–Hurwitz criterion, determine if the system is stable.

E6.4 A control system has the structure shown in Figure E6.4. Determine the gain at which the system will become unstable.

Answer: $K = 20/7$

E6.5 A unity feedback system has a loop transfer function

$$L(s) = \frac{K}{(s + 1)(s + 3)(s + 6)},$$

where $K = 20$. Find the roots of the closed-loop system's characteristic equation.

E6.6 A negative feedback system has a loop transfer function

$$L(s) = G_c(s)G(s) = \frac{K(s + 2)}{s(s - 1)}.$$

(a) Find the value of the gain when the ζ of the closed-loop roots is equal to 0.707. (b) Find the value of the gain when the closed-loop system has two roots on the imaginary axis.

E6.7 For the feedback system of Exercise E6.5, find the value of K when two roots lie on the imaginary axis. Determine the value of the three roots.

Answer: $s = -10, \pm j5.2$

E6.8 Designers have developed small, fast, vertical-take-off fighter aircraft that are invisible to radar (stealth aircraft). This aircraft concept uses quickly turning jet nozzles to steer the airplane [16]. The control system for the heading or direction control is shown in Figure E6.8. Determine the maximum gain of the system for stable operation.

E6.9 A system has a characteristic equation

$$s^3 + 5s^2 + (K + 1)s + 10 = 0.$$

Find the range of K for a stable system.

Answer: $K > 1$

E6.10 Consider a feedback system with closed-loop transfer function

$$T(s) = \frac{4}{s^3 + 4s^2 + s + 4}.$$

Is the system stable?

E6.11 A system with a transfer function $Y(s)/R(s)$ is

$$\frac{Y(s)}{R(s)} = \frac{10(s + 1)}{s^4 + 6s^3 + s^2 + s + 3}.$$

Determine the steady-state error to a unit step input. Is the system stable?

E6.12 A system has the second-order characteristic equation

$$s^2 + as + b = 0,$$

where a and b are constant parameters. Determine the necessary and sufficient conditions for the system to be stable. Is it possible to determine stability of a second-order system just by inspecting the coefficients of the characteristic equation?

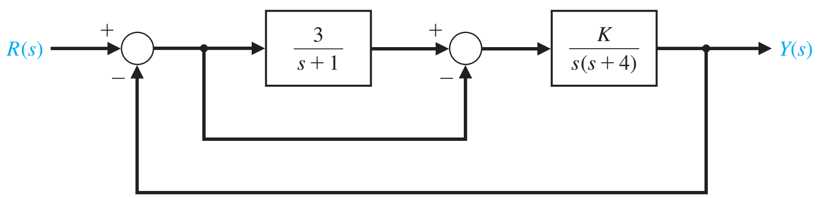


FIGURE E6.4 Feedforward system.

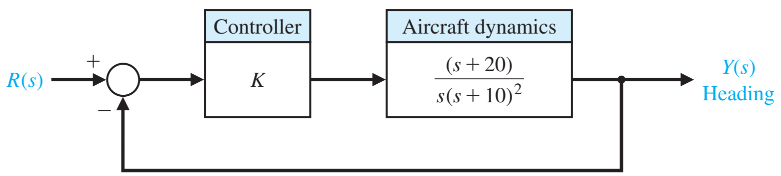


FIGURE E6.8 Aircraft heading control.

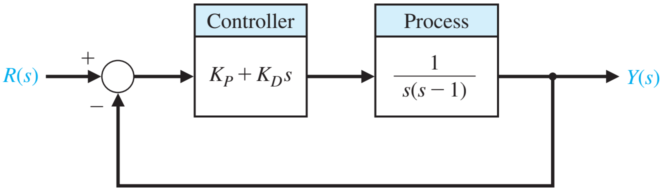


FIGURE E6.13 Closed-loop system with a proportional plus derivative controller $G_c(s) = K_P + K_D s$.

E6.13 Consider the feedback system in Figure E6.13. Determine the range of K_P and K_D for stability of the closed-loop system.

E6.14 By using magnetic bearings, a rotor is supported contactless. The technique of contactless support for rotors becomes more important in light and heavy industrial applications [14]. The matrix differential equation for a magnetic bearing system is

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ -3 & -1 & 0 \\ -2 & -1 & -2 \end{bmatrix} \mathbf{x}(t),$$

where $\mathbf{x}^T(t) = (y(t), \dot{y}(t), i(t))$, $y(t)$ = bearing gap, and $i(t)$ is the electromagnetic current. Determine whether the system is stable.

Answer: The system is stable.

E6.15 A system has a characteristic equation

$$q(s) = s^6 + 9s^5 + 31.25s^4 + 61.25s^3 + 67.75s^2 + 14.75s + 15 = 0.$$

(a) Determine whether the system is stable, using the Routh–Hurwitz criterion. (b) Determine the roots of the characteristic equation.

Answer: (a) The system is marginally stable. (b) $s = -3, -4, -1 \pm 2j, \pm 0.5j$

E6.16 A system has a characteristic equation

$$q(s) = s^4 + 9s^3 + 45s^2 + 87s + 50 = 0.$$

(a) Determine whether the system is stable, using the Routh–Hurwitz criterion. (b) Determine the roots of the characteristic equation.

E6.17 The matrix differential equation of a state variable model of a system is

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & -1 \\ -8 & -12 & 8 \\ -8 & -12 & 5 \end{bmatrix} \mathbf{x}(t).$$

(a) Determine the characteristic equation. (b) Determine whether the system is stable. (c) Determine the roots of the characteristic equation.

Answer: (a) $q(s) = s^3 + 7s^2 + 36s + 24 = 0$

E6.18 A system has a characteristic equation

$$q(s) = s^3 + 20s^2 + 5s + 100 = 0.$$

(a) Determine whether the system is stable, using the Routh–Hurwitz criterion. (b) Determine the roots of the characteristic equation.

E6.19 Determine whether the systems with the following characteristic equations are stable or unstable:
(a) $s^3 + 3s^2 + 5s + 75 = 0$,
(b) $s^4 + 5s^3 + 10s^2 + 10s + 80 = 0$, and
(c) $s^2 + 6s + 3 = 0$.

E6.20 Find the roots of the following polynomials:
(a) $s^3 + 5s^2 + 8s + 4 = 0$ and
(b) $s^3 + 9s^2 + 27s + 27 = 0$.

E6.21 A system has a transfer function $Y(s)/R(s) = T(s) = 1/s$. (a) Is this system stable? (b) If $r(t)$ is a unit step input, determine the response $y(t)$.

E6.22 A system has the characteristic equation

$$q(s) = s^3 + 10s^2 + 29s + K = 0.$$

Shift the vertical axis to the right by 2 by using $s = s_n - 2$, and determine the value of gain K so that the complex roots are $s = -2 \pm j$.

E6.23 The matrix differential equation of a state variable model of a system is

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -k & -4 \end{bmatrix} \mathbf{x}(t).$$

Find the range of k where the system is stable.

E6.24 Consider the system represented in state variable form

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t),$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k & -k & -k \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{C} = [1 \quad 0 \quad 0], \mathbf{D} = [0].$$

(a) What is the system transfer function? (b) For what values of k is the system stable?

E6.25 A closed-loop feedback system is shown in Figure E6.25. For what range of values of the parameters K and p is the system stable?

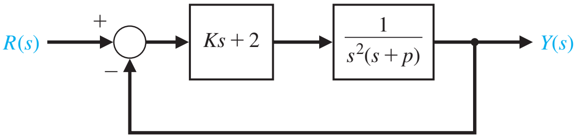


FIGURE E6.25 Closed-loop system with parameters K and p.

E6.26 Consider the closed-loop system in Figure E6.26, where

$$G(s) = \frac{4}{s-1} \quad \text{and} \quad G_c(s) = \frac{1}{2s+K}.$$

(a) Determine the characteristic equation associated with the closed-loop system.
(b) Determine the values of K for which the closed-loop system is stable.

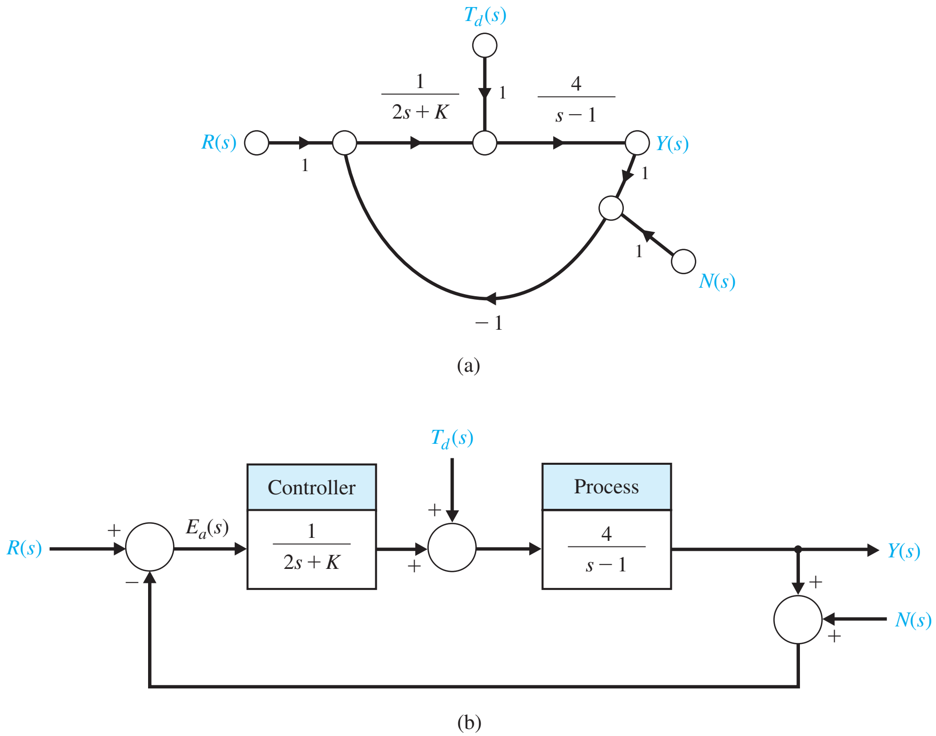


FIGURE E6.26 Closed-loop feedback control system with parameter K.

PROBLEMS

P6.1 Utilizing the Routh–Hurwitz criterion, determine the stability of the following polynomials:

(a) $s^2 + 5s + 2 = 0$
(b) $s^3 + 4s^2 + 8s + 4 = 0$
(c) $s^3 + 2s^2 - 6s + 20 = 0$
(d) $s^4 + s^3 + 2s^2 + 12s + 10 = 0$
(e) $s^4 + s^3 + 3s^2 + 2s + K = 0$
(f) $s^5 + s^4 + 2s^3 + s + 6 = 0$
(g) $s^5 + s^4 + 2s^3 + s^2 + s + K = 0$

Determine the number of roots, if any, in the right-hand plane. If it is adjustable, determine the range of K that results in a stable system.

P6.2 An antenna control system was analyzed in Problem P4.5, and it was determined that, to reduce the effect of wind disturbances, the gain of the magnetic amplifier, k_a, should be as large as possible. (a) Determine the limiting value of gain for maintaining a stable system. (b) We want to have a system settling time equal to 1.5 seconds. Using a shifted axis and the Routh–Hurwitz criterion, determine the value of the gain that satisfies this requirement. Assume that the complex roots of the closed-loop system dominate the transient response. (Is this a valid approximation in this case?)

P6.3 Arc welding is one of the most important areas of application for industrial robots [11]. In most manufacturing welding situations, uncertainties in dimensions of the part, geometry of the joint, and the welding process itself require the use of sensors for maintaining weld quality. Several systems use a vision system to measure the geometry of the puddle of melted metal, as shown in Figure P6.3. This system uses a constant rate of feeding the wire to be melted. (a) Calculate the maximum value for K for the system that will result in a stable system. (b) For half of the maximum value of K found in part (a), determine the roots of the characteristic equation. (c) Estimate the overshoot of the system of part (b) when it is subjected to a step input.

P6.4 A feedback control system is shown in Figure P6.4. The controller and process transfer functions are given by

$$G_c(s) = K \text{ and } G(s) = \frac{s + 40}{s(s + 10)}$$

and the feedback transfer function is $H(s) = 1/(s + 20)$. (a) Determine the limiting value of gain K for a stable system. (b) For the gain that results in marginal stability, determine the magnitude of the imaginary roots. (c) Reduce the gain to half the magnitude of the marginal value and determine the relative stability of the system (1) by shifting the axis and using the Routh–Hurwitz criterion and (2) by determining the root locations. Show the roots are between -1 and -2.

P6.5 Determine the relative stability of the systems with the following characteristic equations (1) by shifting the axis in the s-plane and using the Routh–Hurwitz criterion, and (2) by determining the location of the complex roots in the s-plane:

(a) $s^3 + 5s^2 + 6s + 2 = 0$.

(b) $s^4 + 9s^3 + 30s^2 + 42s + 20 = 0$.

(c) $s^3 + 20s^2 + 100s + 200 = 0$.

P6.6 A unity-feedback control system is shown in Figure P6.6. Determine the relative stability of the

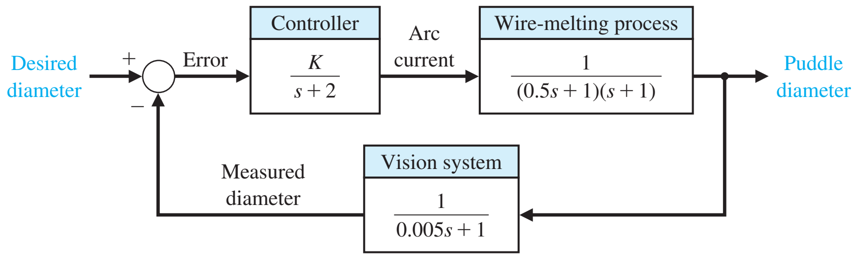


FIGURE P6.3
Welder control.

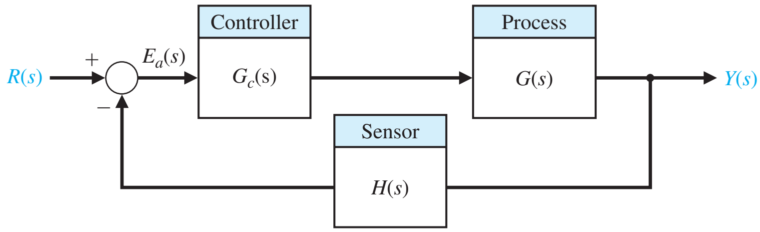


FIGURE P6.4
Nonunity feedback system.

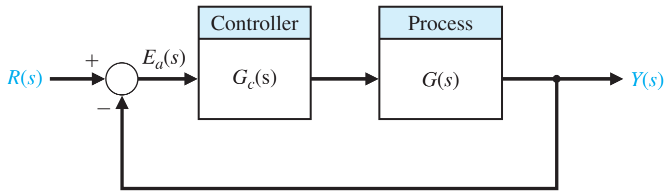


FIGURE P6.6
Unity feedback system.

system with the following loop transfer functions by locating the complex roots in the s-plane:

(a) $G_c(s)G(s) = \dfrac{10s + 2}{s^2(s + 1)}$

(b) $G_c(s)G(s) = \dfrac{24}{s(s^3 + 10s^2 + 35s + 50)}$

(c) $G_c(s)G(s) = \dfrac{(s + 2)(s + 3)}{s(s + 4)(s + 6)}$

P6.7 The linear model of a phase detector (phase-lock loop) can be represented by Figure P6.7 [9]. The phase-lock systems are designed to maintain zero difference in phase between the input carrier signal and a local voltage-controlled oscillator. The filter for a particular application is chosen as

$$F(s) = \frac{10(s + 10)}{(s + 1)(s + 100)}.$$

We want to minimize the steady-state error of the system for a ramp change in the phase information signal. (a) Determine the limiting value of the gain $K_aK = K_v$ in order to maintain a stable system. (b) A steady-state error equal to 1° is acceptable for a ramp signal of 100 rad/s. For that value of gain K_v, determine the location of the roots of the system.

P6.8 A very interesting and useful velocity control system has been designed for a wheelchair control system. A proposed system utilizing velocity sensors mounted in a headgear is shown in Figure P6.8. The headgear sensor provides an output proportional to the magnitude of the head movement. There is a sensor mounted at 90° intervals so that forward, left, right, or reverse can be commanded. Typical values for the time constants are $\tau_1 = 0.5$ s, $\tau_2 = 1$ s, and $\tau_3 = \frac{1}{4}$ s.

(a) Determine the limiting gain $K = K_1K_2K_3$ for a stable system.

(b) When the gain K is set equal to one-third of the limiting value, determine whether the settling time (to within 2% of the final value of the system) is $T_s \leq 4$ s.

(c) Determine the value of gain that results in a system with a settling time of $T_s \leq 4$ s. Also, obtain the value of the roots of the characteristic equation when the settling time is $T_s \leq 4$ s.

P6.9 A cassette tape storage device has been designed for mass-storage [1]. It is necessary to control the velocity of the tape accurately. The speed control of the tape drive is represented by the system shown in Figure P6.9.

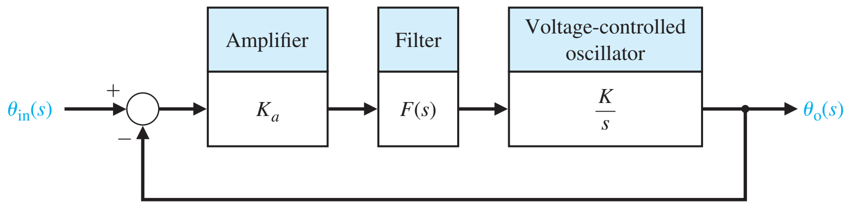


FIGURE P6.7 Phase-lock loop system.

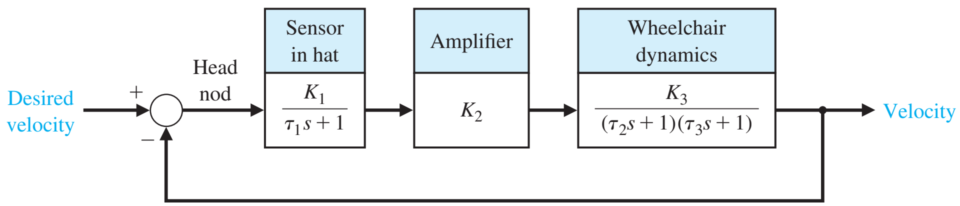


FIGURE P6.8 Wheelchair control system.

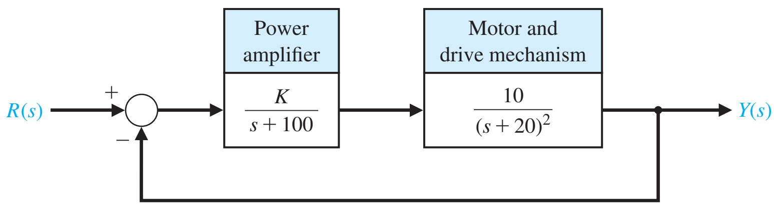


FIGURE P6.9 Tape drive control.

(a) Determine the limiting gain for a stable system.
(b) Determine a suitable gain so that the percent overshoot to a step command is $P.O. = 5\%$.

P6.10 Robots can be used in manufacturing and assembly operations that require accurate, fast, and versatile manipulation [10, 11]. The loop transfer function of a direct-drive arm is

$$G_c(s)G(s) = \frac{K(s + 10)}{s(s + 3)(s^2 + 4s + 8)}.$$

(a) Determine the value of gain K when the system oscillates. (b) Calculate the roots of the closed-loop system for the K determined in part (a).

P6.11 A feedback control system has a characteristic equation

$$s^3 + (1 + K)s^2 + 3s + (1 + 7K) = 0.$$

The parameter K must be positive. What is the maximum value K can assume before the system becomes unstable? When K is equal to the maximum value, the system oscillates. Determine the frequency of oscillation.

P6.12 A system has the third-order characteristic equation

$$s^3 + as^2 + bs + c = 0,$$

where a, b, and c are constant parameters. Determine the necessary and sufficient conditions for the system to be stable. Is it possible to determine stability of the system by just inspecting the coefficients of the characteristic equation?

P6.13 Consider the system in Figure P6.13. Determine the conditions on K, p, and z that must be satisfied for closed-loop stability. Assume that $K > 0$, $\zeta > 0$, and $\omega_n > 0$.

P6.14 A feedback control system has a characteristic equation

$$s^6 + 2s^5 + 13s^4 + 16s^3 + 56s^2 + 32s + 80 = 0.$$

Determine whether the system is stable, and determine the values of the roots.

P6.15 The stability of a motorcycle and rider is an important area for study [12, 13]. The handling characteristics of a motorcycle must include a model of the rider as well as one of the vehicle. The dynamics of one motorcycle and rider can be represented by the loop transfer function

$$L(s) = \frac{K(s^2 + 30s + 1125)}{s(s + 20)(s^2 + 10s + 125)(s^2 + 60s + 3400)}.$$

(a) As an approximation, calculate the acceptable range of K for a stable unity negative feedback system when the numerator polynomial (zeros) and the denominator polynomial $(s^2 + 60s + 3400)$ are neglected. (b) Calculate the actual range of acceptable K, account for all zeros and poles.

P6.16 A system has a closed-loop transfer function

$$T(s) = \frac{1}{s^3 + 5s^2 + 20s + 6}.$$

(a) Determine whether the system is stable. (b) Determine the roots of the characteristic equation. (c) Plot the response of the system to a unit step input.

P6.17 The elevator in Yokohama's 70-story Landmark Tower operates at a peak speed of 45 km/hr. To reach such a speed without inducing discomfort in passengers, the elevator accelerates for longer periods, rather than more precipitously. Going up, it reaches full speed only at the 27th floor; it begins decelerating 15 floors later. The result is a peak acceleration similar to that of other skyscraper elevators—a bit less than a tenth of the force of gravity. Admirable ingenuity has gone into making this safe and comfortable. Special ceramic brakes had to be developed; iron ones would melt. Computer-controlled systems damp out vibrations. The lift has been streamlined to reduce the wind noise as it speeds up and down [19]. One proposed control system for the elevator vertical position is shown in Figure P6.17. Determine the range of K for a stable system.

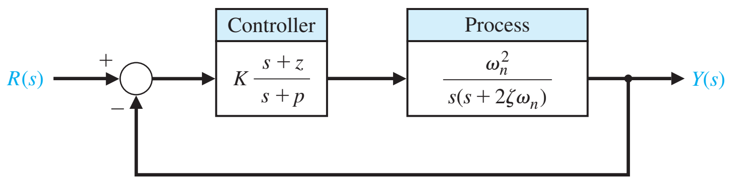


FIGURE P6.13 Control system with controller with three parameters K, p, and z.

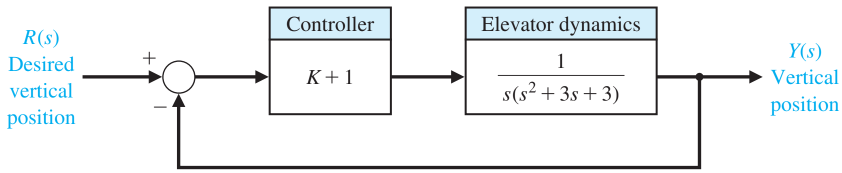


FIGURE P6.17 Elevator control system.

P6.18 Consider the case of rabbits and foxes. The number of rabbits is x_1 and, if left alone, it would grow indefinitely (until the food supply was exhausted) so that

$$\dot{x}_1 = kx_1.$$

However, with foxes present, we have

$$\dot{x}_1 = kx_1 - ax_2,$$

where x_2 is the number of foxes. Now, if the foxes must have rabbits to exist, we have

$$\dot{x}_2 = -hx_2 + bx_1.$$

Determine whether this system is stable and thus decays to the condition $x_1(t) = x_2(t) = 0$ at $t = \infty$. What are the requirements on a, b, h, and k for a stable system? What is the result when k is greater than h?

P6.19 The goal of vertical takeoff and landing (VTOL) aircraft is to achieve operation from relatively small airports and yet operate as a normal aircraft in level flight [16]. An aircraft taking off in a form similar to a rocket is inherently unstable. A control system using adjustable jets can control the vehicle, as shown in Figure P6.19. (a) Determine the range of gain for which the system is stable. (b) Determine the gain K for which the system is marginally stable and the roots of the characteristic equation for this value of K.

P6.20 A personal vertical take-off and landing (VTOL) aircraft is shown in Figure P6.20(a). A possible control system for aircraft altitude is shown in Figure P6.20(b). (a) For $K = 6$, determine whether the system is stable. (b) Determine a range of stability, if any, for $K > 0$.

P6.21 Consider the system described in state variable form by

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}\mathbf{x}(t)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -k_1 & -k_2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \text{ and } \mathbf{C} = [1 \quad -1],$$

and where $k_1 \neq k_2$ and both k_1 and k_2 are real numbers.

(a) Compute the state transition matrix $\Phi(t, 0)$.
(b) Compute the eigenvalues of the system matrix $\mathbf{A}$.
(c) Compute the roots of the characteristic polynomial. (d) Discuss the results of parts (a)–(c) in terms of stability of the system.

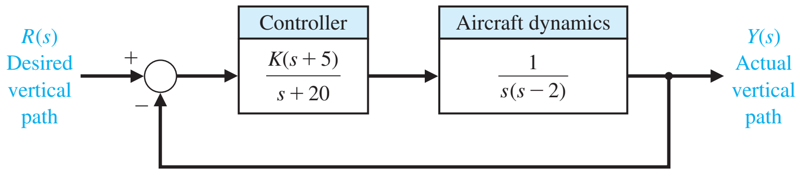


FIGURE P6.19 Control of a jump-jet aircraft.

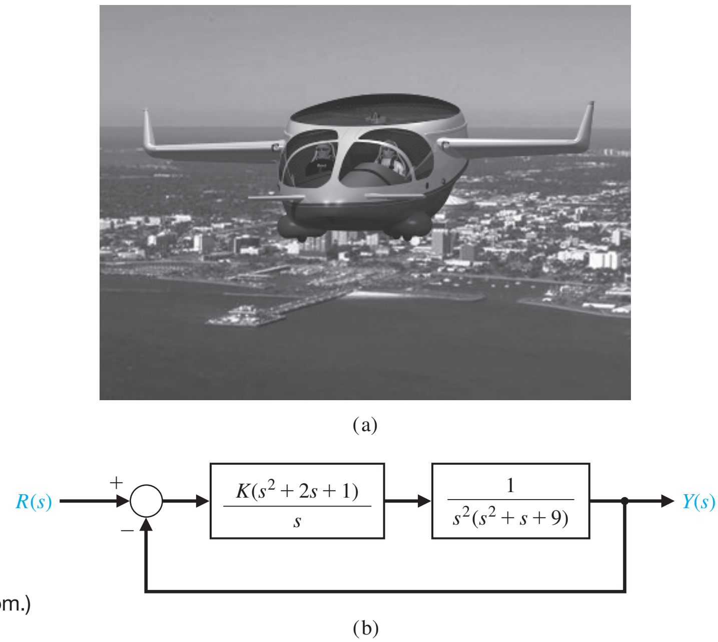


FIGURE P6.20 (a) Personal VTOL aircraft. (Courtesy of Mirror Image Aerospace at www.skywalkervtol.com.) (b) Control system.

ADVANCED PROBLEMS

AP6.1 A teleoperated control system incorporates both a person (operator) and a remote machine. The normal teleoperation system is based on a one-way link to the machine and limited feedback to the operator. However, two-way coupling using bilateral information exchange enables better operation [18]. In the case of remote control of a robot, force feedback plus position feedback is useful. The characteristic equation for a teleoperated system, as shown in Figure AP6.1, is

$$s^4 + 20s^3 + K_1s^2 + 4s + K_2 = 0,$$

where K_1 and K_2 are feedback gain factors. Determine and plot the region of stability for this system for K_1 and K_2.

AP6.2 Consider the case of a navy pilot landing an aircraft on an aircraft carrier. The pilot has three basic tasks. The first task is guiding the aircraft's approach to the ship along the extended centerline of the runway. The second task is maintaining the aircraft on the correct glideslope. The third task is maintaining the correct speed. A model of a lateral position control system is shown in Figure AP6.2. Determine the range of stability for $K \geq 0$.

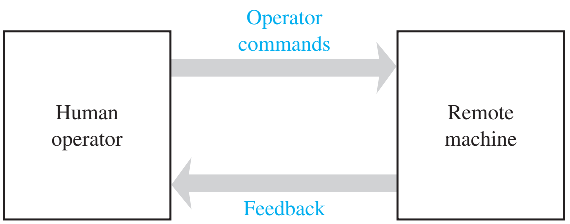


FIGURE AP6.1 Model of a teleoperated machine.

AP6.3 A control system is shown in Figure AP6.3. We want the system to be stable and the steady-state error for a unit step input to be less than or equal to 0.05. (a) Determine the range of α that satisfies the error requirement. (b) Determine the range of α that satisfies the stability requirement. (c) Select an α that meets both requirements.

AP6.4 A bottle-filling line uses a feeder screw mechanism, as shown in Figure AP6.4. The tachometer feedback is used to maintain accurate speed control. Determine and plot the range of K and p that permits stable operation.

AP6.5 Consider the closed-loop system in Figure AP6.5. Suppose that all gains are positive, that is, $K_1 > 0$, $K_2 > 0$, $K_3 > 0$, $K_4 > 0$, and $K_5 > 0$.

(a) Determine the closed-loop transfer function $T(s) = Y(s)/R(s)$.

(b) Obtain the conditions on selecting the gains K_1, K_2, K_3, K_4, and K_5, so that the closed-loop system is guaranteed to be stable.

(c) Using the results of part (b), select values of the five gains so that the closed-loop system is stable, and plot the unit step response.

AP6.6 A spacecraft with a camera is shown in Figure AP6.6(a). The camera slews about 16° in a canted plane relative to the base. Reaction jets stabilize the base against the reaction torques from the slewing motors. Suppose that the rotational speed control for the camera slewing has a plant transfer function

$$G(s) = \frac{1}{(s + 2)(s + 3)(s + 4)}.$$

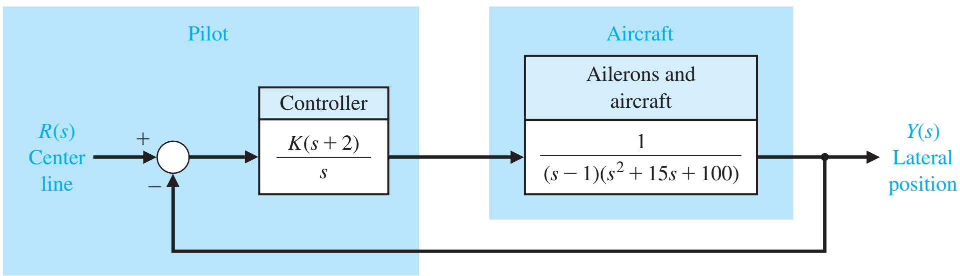


FIGURE AP6.2 Lateral position control for landing on an aircraft carrier.

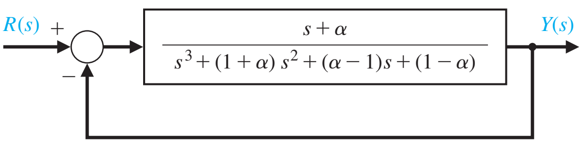
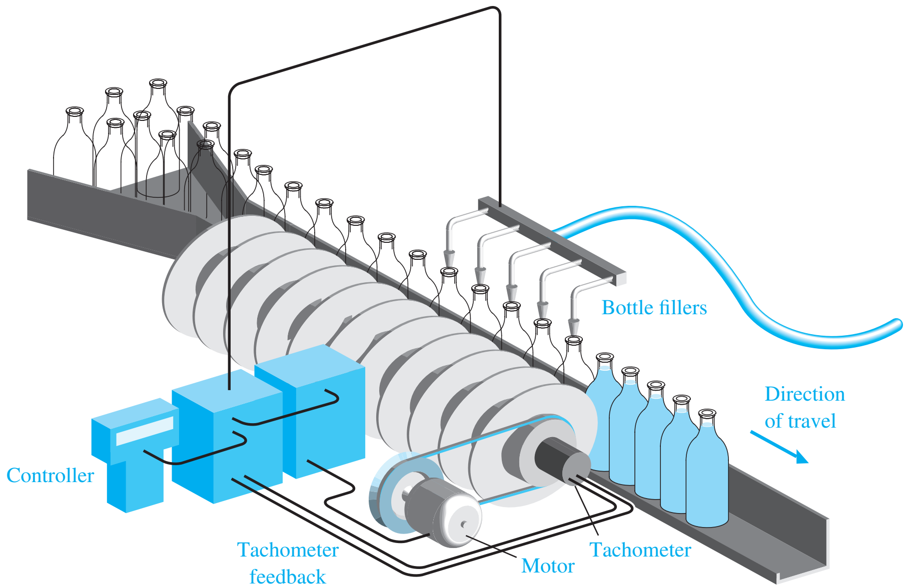
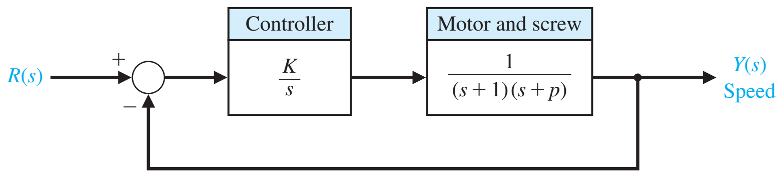


FIGURE AP6.3 Third-order unity feedback system.

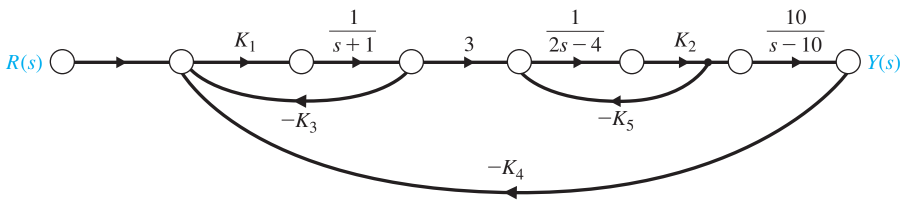


(a)

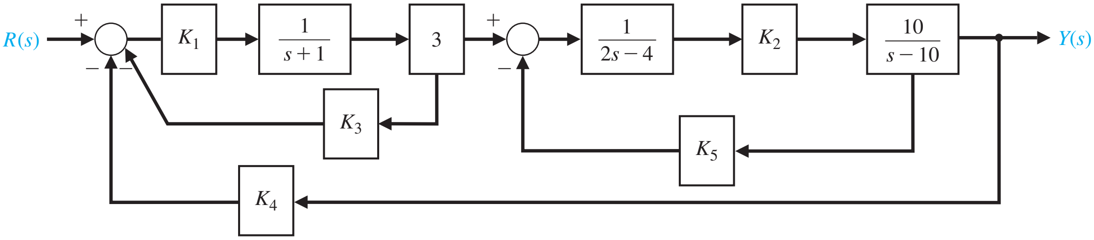


(b)

FIGURE AP6.4 Speed control of a bottle-filling line. (a) System layout. (b) Block diagram.



(a)



(b)

FIGURE AP6.5 Multiloop feedback control system. (a) Signal flow graph. (b) Block diagram.

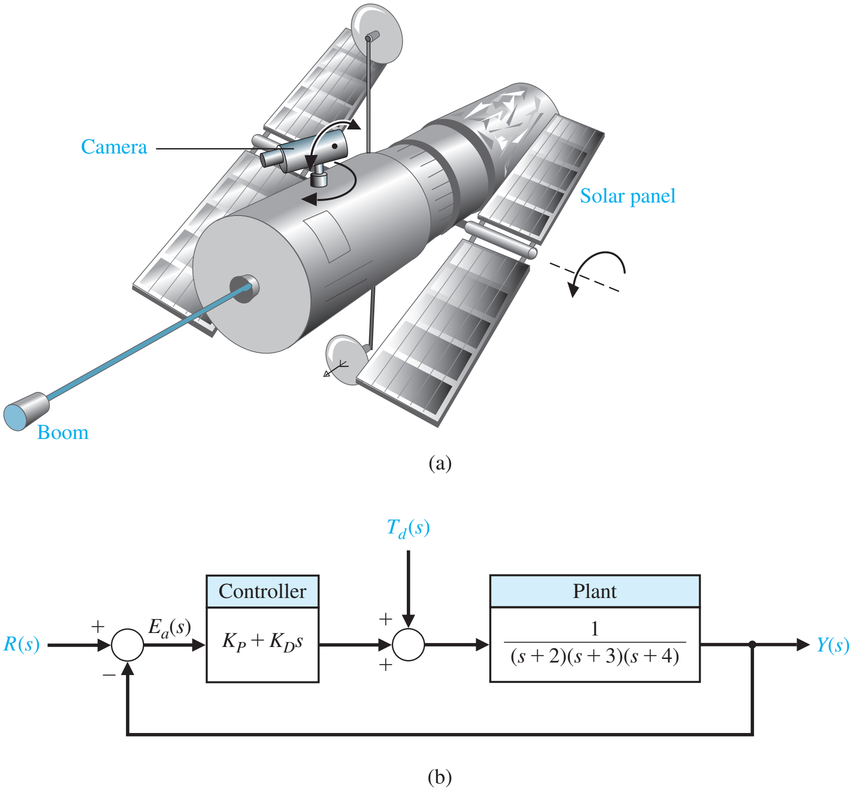


FIGURE AP6.6 (a) Spacecraft with a camera. (b) Feedback control system.

A proportional plus derivative controller is used in a system as shown in Figure AP6.6(b), where

$$G_c(s) = K_p + K_D s,$$

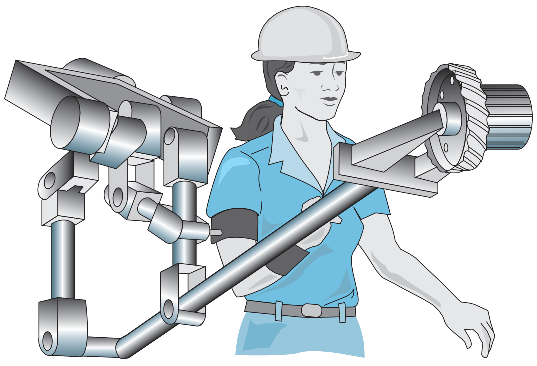
and where $K_p > 0$ and $K_D > 0$. Obtain and plot the relationship between K_p and K_D that results in a stable closed-loop system.

AP6.7 A human's ability to perform physical tasks is limited not by intellect but by physical strength. If, in an appropriate environment, a machine's mechanical power is closely integrated with a human arm's mechanical strength under the control of the human intellect, the resulting system will be superior to a loosely integrated combination of a human and a fully automated robot.

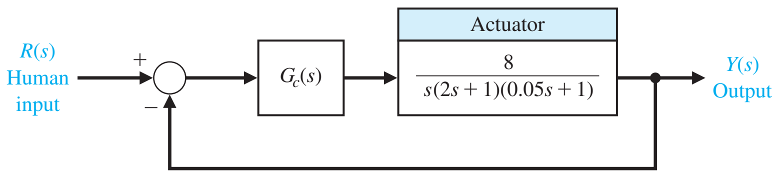
Extenders are defined as a class of robot manipulators that extend the strength of the human arm while maintaining human control of the task [23]. The defining characteristic of an extender is the transmission of both power and information signals. The extender is worn by the human; the physical contact between the extender and the human allows the direct transfer of mechanical power and information signals. Because of this unique interface, control of the extender trajectory can be accomplished without any type of joystick, keyboard, or master-slave system. The human provides a control system for the extender, while the extender actuators provide most of the strength necessary for the task. The human becomes a part of the extender and "feels" a scaled-down version of the load that the extender is carrying. The extender is distinguished from a conventional master-slave system; in that type of system, the human operator is either at a remote location or close to the slave manipulator, but is not in direct physical contact with the slave in the sense of transfer of power. An extender is shown in Figure AP6.7(a) [23]. The block diagram of the system is shown in Figure AP6.7(b). Consider the proportional plus integral controller

$$G_c(s) = K_P + \frac{K_I}{s}.$$

Determine the range of values of the controller gains K_P and K_I such that the closed-loop system is stable.



(a)



(b)

FIGURE AP6.7 Extender robot control.

DESIGN PROBLEMS

CDP6.1 The capstan drive system of problem CDP5.1 uses the amplifier as the controller. Determine the maximum value of the gain K_a before the system becomes unstable.

DP6.1 The control of the spark ignition of an automotive engine requires constant performance over a wide range of parameters [15]. The control system is shown in Figure DP6.1, with a controller gain K to be selected. The parameter p is equal to 2 for many autos but can equal zero for those with high performance. Select a gain K that will result in a stable system for both values of p.

DP6.2 An automatically guided vehicle on Mars is represented by the system in Figure DP6.2. The system has a steerable wheel in both the front and back of the vehicle, and the design requires that $H(s) = Ks + 1$. Determine (a) the value of K required for stability, (b) the value of K when one root of the characteristic equation is equal to $s = -5$, and (c) the value of the two remaining roots for the gain selected in

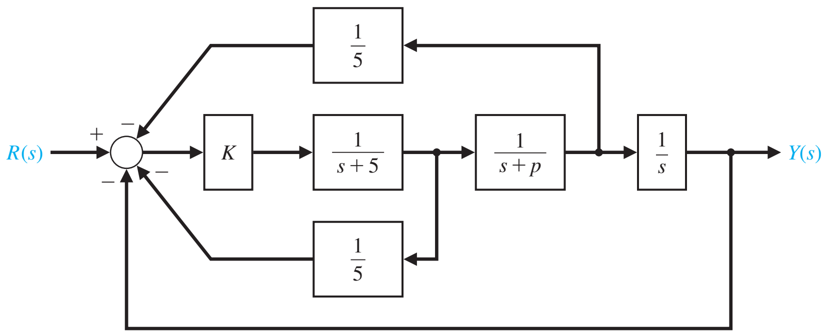


FIGURE DP6.1 Automobile engine control.

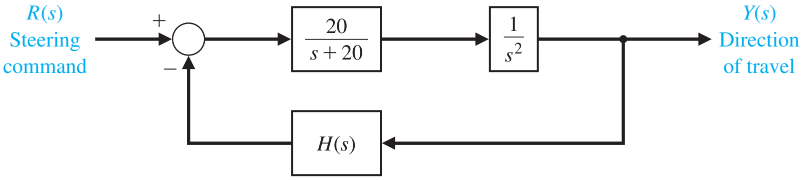


FIGURE DP6.2 Mars guided vehicle control.

part (b). (d) Find the response of the system to a step command for the gain selected in part (b).

DP6.3 A unity negative feedback system with

$$L(s) = G_c(s)G(s) = \frac{K(s + 2)}{s(1 + \tau s)(1 + 2s)}$$

has two parameters to be selected. (a) Determine and plot the regions of stability for this system. (b) Select τ and K so that the steady-state error to a ramp input is less than or equal to 25% of the input magnitude. (c) Determine the percent overshoot for a step input for the design selected in part (b).

DP6.4 The attitude control system of a rocket is shown in Figure DP6.4 [17]. (a) Determine the range of gain K and parameter m so that the system is stable, and plot the region of stability. (b) Select the gain and parameter values so that the steady-state error to a ramp input is less than or equal to 10% of the input magnitude. (c) Determine the percent overshoot for a step input for the design selected in part (b).

DP6.5 A traffic control system is designed to control the distance between vehicles, as shown in Figure DP6.5 [15]. (a) Determine the range of gain K for which the system is stable. (b) If K_m is the maximum value of K so that the characteristic roots are on the $j\omega$-axis, then let $K = K_m/N$, where $6 < N < 7$. We want the peak time to be $T_p \le 2$ s and the percent overshoot to be $P.O. \le 18\%$. Determine an appropriate value for N.

DP6.6 Consider the single-input, single-output system as described by

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}\mathbf{x}(t)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 2 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{C} = [1 \quad 0].$$

Assume that the input is a linear combination of the states, that is,

$$u(t) = -\mathbf{K}\mathbf{x}(t) + r(t),$$

where $r(t)$ is the reference input. The matrix $\mathbf{K} = [K_1 \quad K_2]$ is known as the gain matrix. If you substitute $u(t)$ into the state variable equation you obtain the closed-loop system

$$\dot{\mathbf{x}}(t) = [\mathbf{A} - \mathbf{B}\mathbf{K}]\mathbf{x}(t) + \mathbf{B}r(t)$$
$$y(t) = \mathbf{C}\mathbf{x}(t)$$

For what values of $\mathbf{K}$ is the closed-loop system stable? Determine the region of the left half-plane where the desired closed-loop eigenvalues should be placed

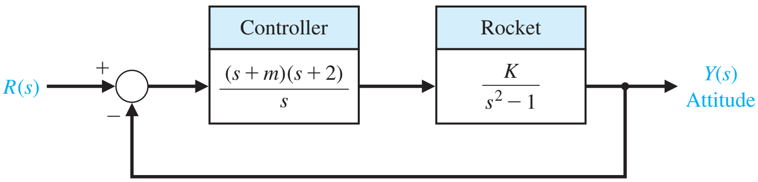


FIGURE DP6.4 Rocket attitude control.

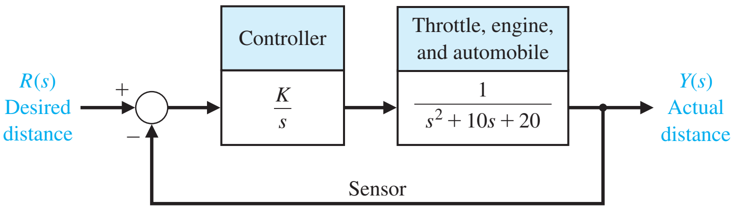


FIGURE DP6.5 Traffic distance control.

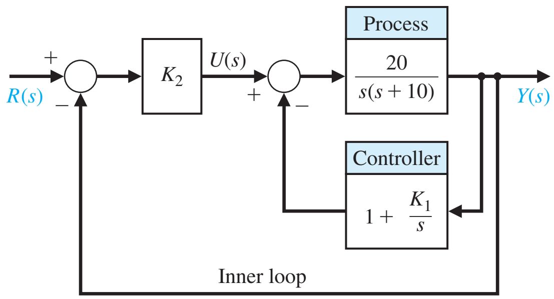


FIGURE DP6.7 Feedback system with inner and outer loop.

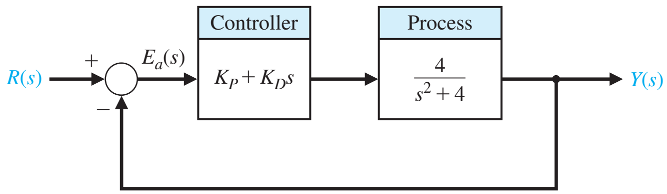


FIGURE DP6.8 A marginally stable plant with a PD controller in the loop.

so that the percent overshoot to a unit step input, $R(s) = 1/s$, is $P.O. < 5\%$ and the settling time is $T_s < 4s$. Select a gain matrix, $\mathbf{K}$, so that the system step response meets the specifications $P.O. < 5\%$ and $T_s < 4s$.

DP6.7 Consider the feedback control system in Figure DP6.7. The system has an inner loop and an outer loop. The inner loop must be stable and have a quick speed of response. (a) Consider the inner loop first. Determine the range of K_1 resulting in a stable inner loop. That is, the transfer function $Y(s)/U(s)$ must be stable. (b) Select the value of K_1 in the stable range leading to the fastest step response. (c) For the value of K_1 selected in (b), determine the range of K_2 such that the closed-loop system $T(s) = Y(s)/R(s)$ is stable.

DP6.8 Consider the feedback system shown in Figure DP6.8. The process transfer function is marginally stable. The controller is the proportional-derivative (PD) controller

$$G_c(s) = K_P + K_D s.$$

Determine if it is possible to find values of K_P and K_D such that the closed-loop system is stable. If so, obtain values of the controller parameters such that the steady-state tracking error $E(s) = R(s) - Y(s)$ to a unit step input $R(s) = 1/s$ is $e_{ss} = \lim_{t\to\infty} e(t) \le 0.1$ and the damping of the closed-loop system is $\zeta = \sqrt{2}/2$.

COMPUTER PROBLEMS

CP6.1 Determine the roots of the following characteristic equations:

(a) $q(s) = s^3 + 2s^2 + 20s + 10 = 0.$

(b) $q(s) = s^4 + 8s^3 + 24s^2 + 32s + 16 = 0.$

(c) $q(s) = s^4 + 2s^2 + 1 = 0.$

CP6.2 Consider a unity negative feedback system with

$$G_c(s) = K \text{ and } G(s) = \frac{s^2 - 2s + 4}{s^2 + 4s + 2}.$$

Develop an m-file to compute the roots of the closed-loop transfer function characteristic polynomial for $K = 1, 2,$ and 5. For which values of K is the closed-loop system stable?

CP6.3 A unity negative feedback system has the loop transfer function

$$L(s) = G_c(s)G(s) = \frac{s + 1}{s^3 + 4s^2 + 6s + 10}.$$

Develop an m-file to determine the closed-loop transfer function and show that the roots of the characteristic equation are $s_1 = -2.89$ and $s_{2,3} = -0.55 \pm j1.87$.

CP6.4 Consider the closed-loop transfer function

$$T(s) = \frac{1}{s^5 + 2s^4 + 2s^3 + 4s^2 + s + 2}.$$

(a) Using the Routh–Hurwitz method, determine whether the system is stable. If it is not stable, how many poles are in the right half-plane? (b) Compute the poles of $T(s)$ and verify the result in part (a). (c) Plot the unit step response, and discuss the results.

CP6.5 A "paper-pilot" model is sometimes utilized in aircraft control design and analysis to represent the pilot in the loop. A block diagram of an aircraft with a pilot "in the loop" is shown in Figure CP6.5. The variable τ represents the pilot's time delay. We can represent a slower pilot with $\tau = 0.6$ and a faster pilot with $\tau = 0.1$. The remaining variables in the pilot model are assumed to be $K = 1$, $\tau_1 = 2$, and $\tau_2 = 0.5$. Develop an m-file to compute the closed-loop system poles for the fast and slow pilots. Comment on the results. What is the maximum pilot time delay allowable for stability?

CP6.6 Consider the feedback control system in Figure CP6.6. Using the for function, develop an m-file script to compute the closed-loop transfer function poles for $0 \le K \le 5$ and plot the results denoting the poles with the "×" symbol. Determine the maximum range of K for stability with the Routh–Hurwitz method. Compute the roots of the characteristic equation when K is the minimum value allowed for stability.

CP6.7 Consider a system in state variable form:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -12 & -14 & -10 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix} u(t),$$

$$y(t) = [1 \quad 1 \quad 0]\mathbf{x}(t).$$

(a) Compute the characteristic equation using the poly function. (b) Compute the roots of the characteristic equation, and determine whether the system is stable. (c) Obtain the response plot of $y(t)$ when $u(t)$ is a unit step and when the system has zero initial conditions.

CP6.8 Consider the feedback control system in Figure CP6.8. (a) Using the Routh–Hurwitz method, determine the range of K_1 resulting in closed-loop stability. (b) Develop an m-file to plot the pole locations as a function of $0 < K_1 < 30$ and comment on the results.

CP6.9 Consider a system represented in state variable form

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

$$y(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t),$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ -k & -3 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{C} = [1 \quad 2 \quad 0], \mathbf{D} = [0].$$

(a) For what values of k is the system stable?

(b) Develop an m-file to plot the pole locations as a function of $0 < k < 10$ and comment on the results.

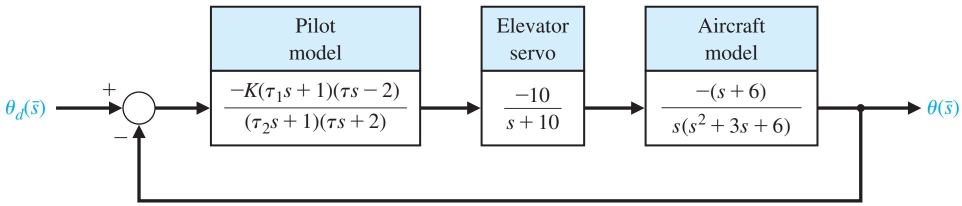


FIGURE CP6.5 An aircraft with a pilot in the loop.

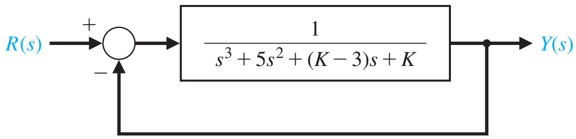


FIGURE CP6.6 A single-loop feedback control system with parameter K.

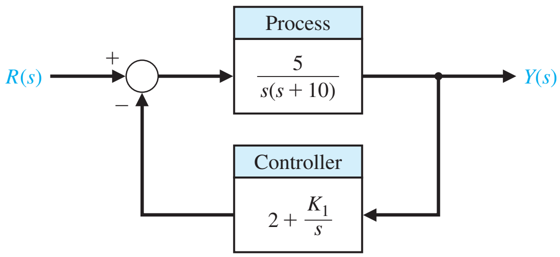


FIGURE CP6.8 Nonunity feedback system with parameter K_1.

ANSWERS TO SKILLS CHECK

True or False: (1) False; (2) True; (3) False; (4) True; (5) True
Multiple Choice: (6) a; (7) c; (8) a; (9) b; (10) b; (11) a; (12) a; (13) b; (14) a; (15) b
Word Match (in order, top to bottom): e, d, f, a, b, g, c

TERMS AND CONCEPTS

Absolute stability A system description that reveals whether a system is stable or not stable without consideration of other system attributes such as degree of stability.

Auxiliary polynomial The equation that immediately precedes the zero entry in the Routh array.

Marginally stable A system is marginally stable if and only if the zero input response remains bounded as $t \rightarrow \infty$.

Relative stability The property that is measured by the relative real part of each root or pair of roots of the characteristic equation.

Routh–Hurwitz criterion A criterion for determining the stability of a system by examining the characteristic equation of the transfer function. The criterion states that the number of roots of the characteristic equation with positive real parts is equal to the number of changes of sign of the coefficients in the first column of the Routh array.

Stability A performance measure of a system. A system is stable if all the poles of the transfer function have negative real parts.

Stable system A dynamic system with a bounded system response to a bounded input.